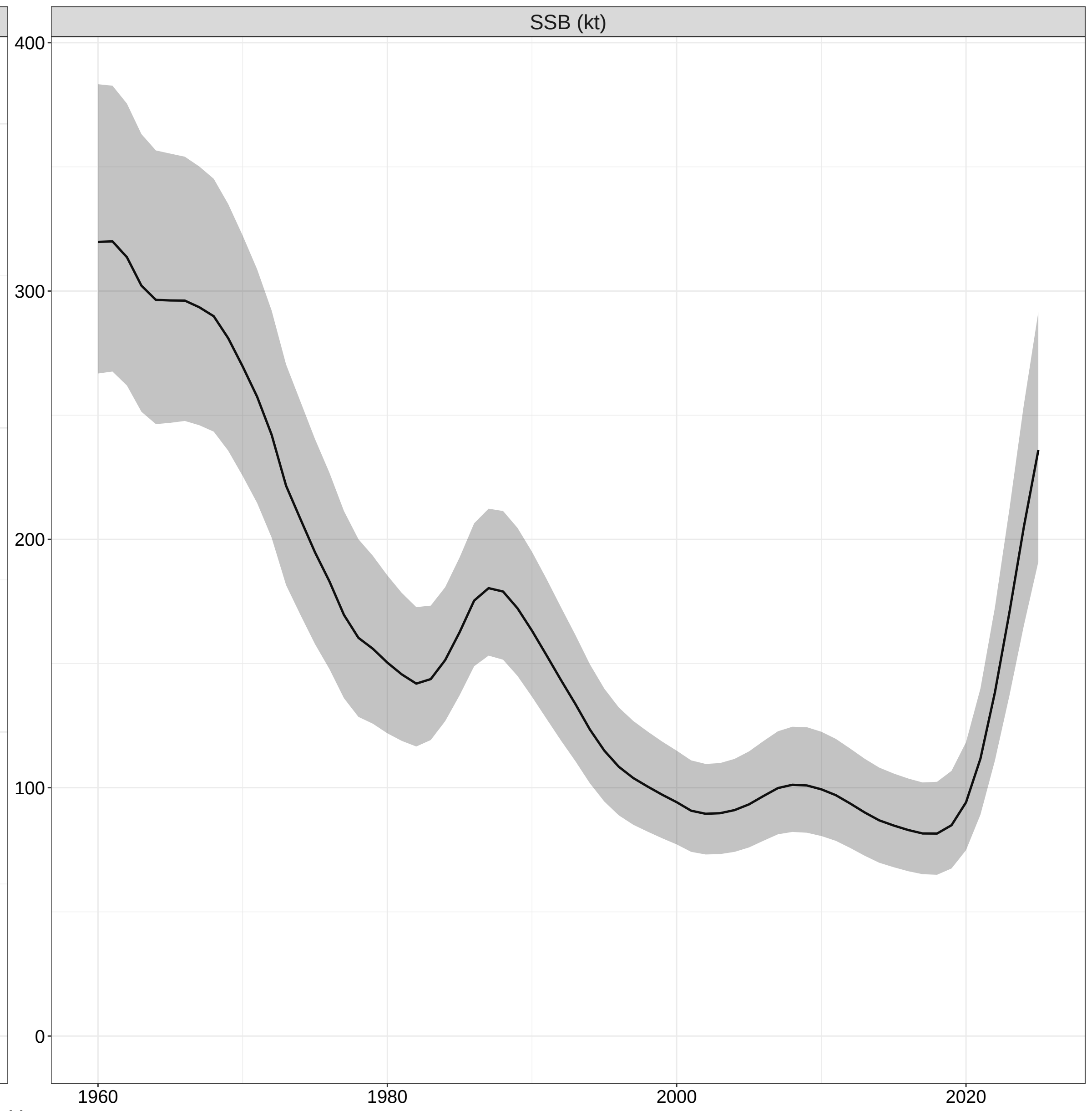
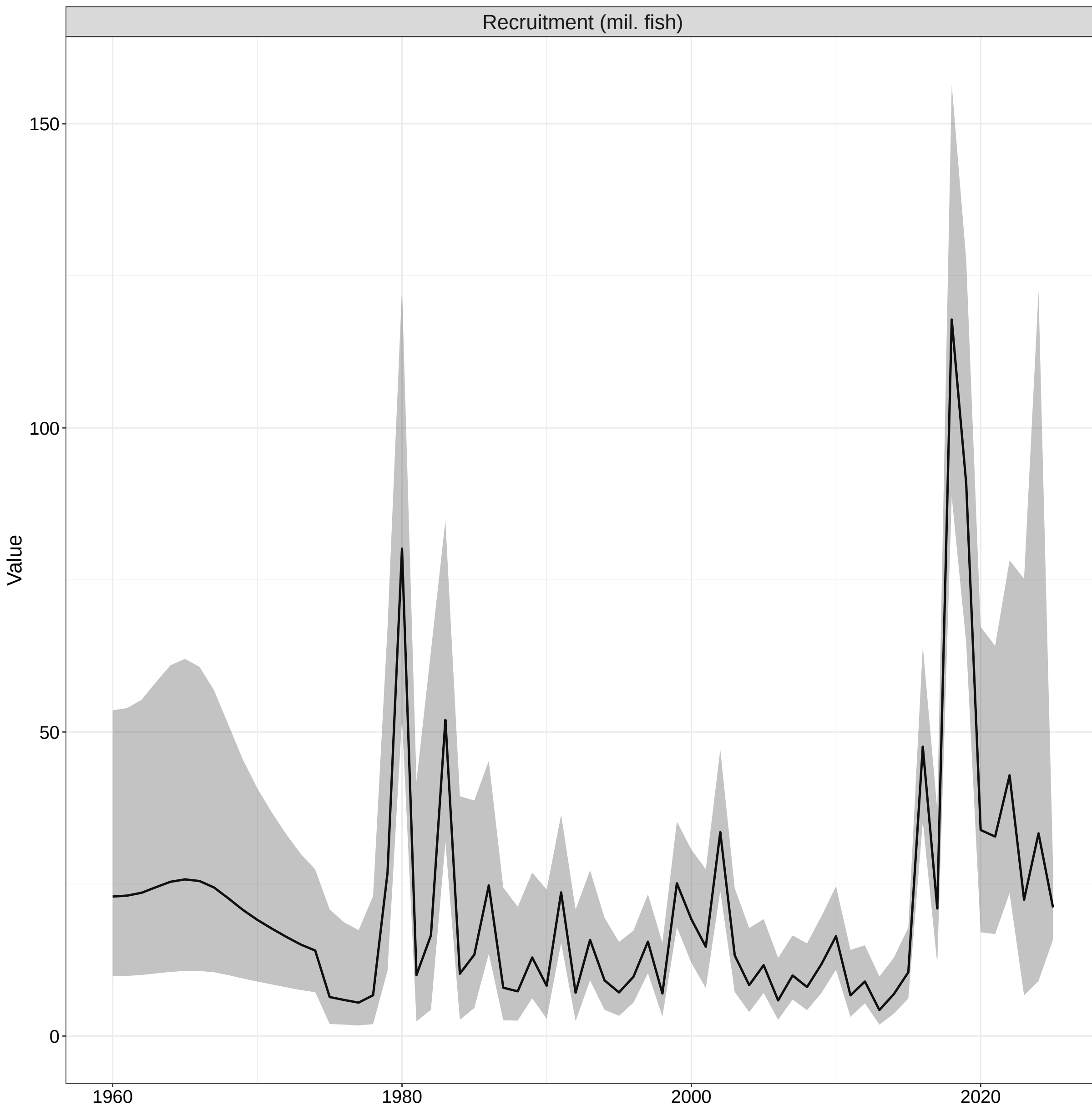
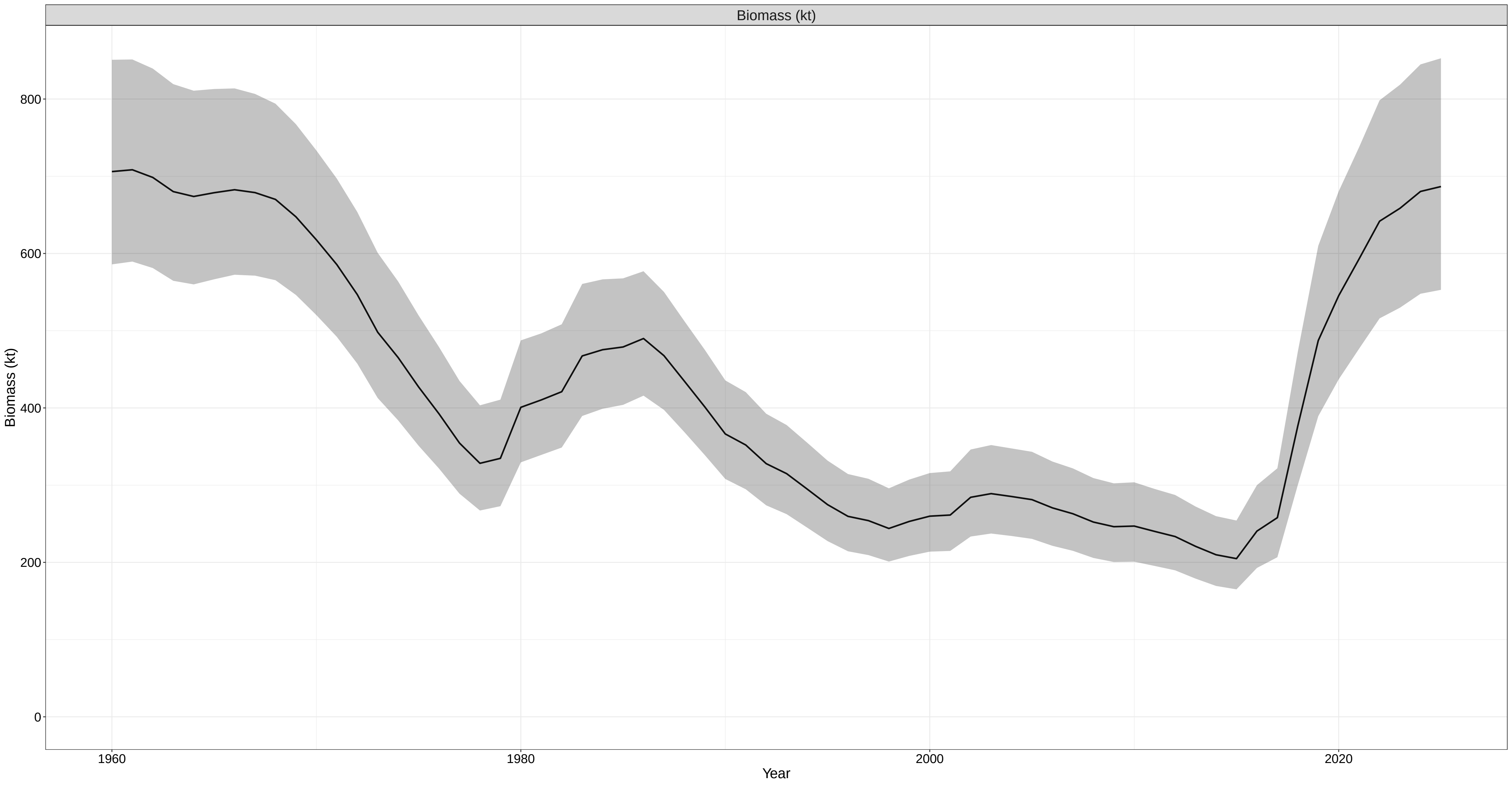


Source	SE	Eff_SE	Data_Wt	ISS	ESS
LL Survey Index	0.07	0.07	1	NA	NA
JPN LL Survey Index	0.07	0.07	1	NA	NA
Fixed Gear Fish Ages	NA	NA	1	145.15	145.15
LL Survey Ages	NA	NA	1	171.43	171.43
JPN LL Survey Ages	NA	NA	1	83.33	83.33
Trawl Fish Lengths	NA	NA	1	45.39	45.39
JPN LL Survey Lengths	NA	NA	1	54.62	54.62

Correlation of 0.950860784563274. This indicates potential non-convergence according to the tolerance.

	Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
1	1	1	235.94577	387.13814	0.0444656656112626	44.12395	141.14731	0.08213	1.67163	0.60946	0.54143

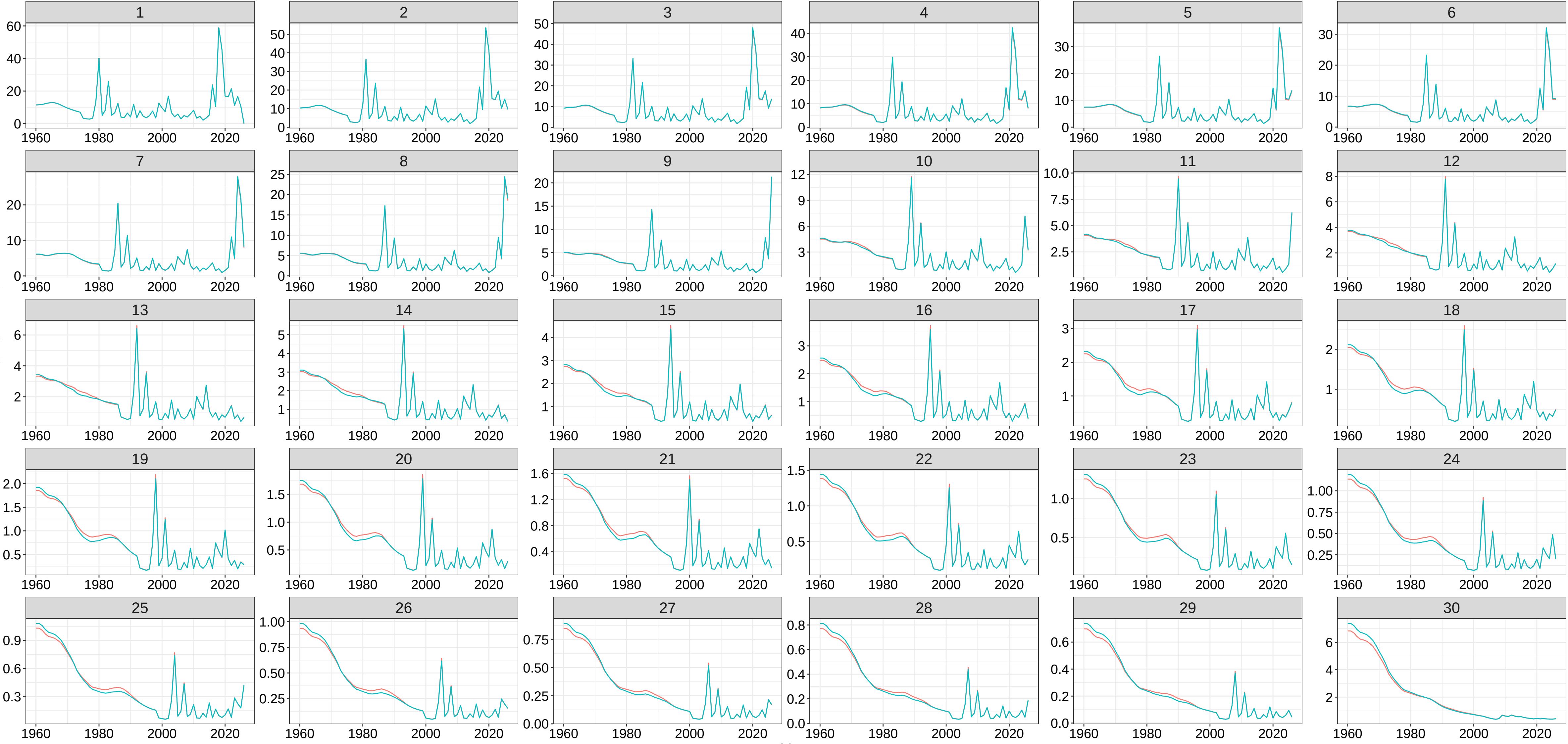




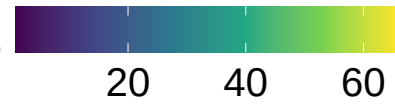
Numbers-at-Age

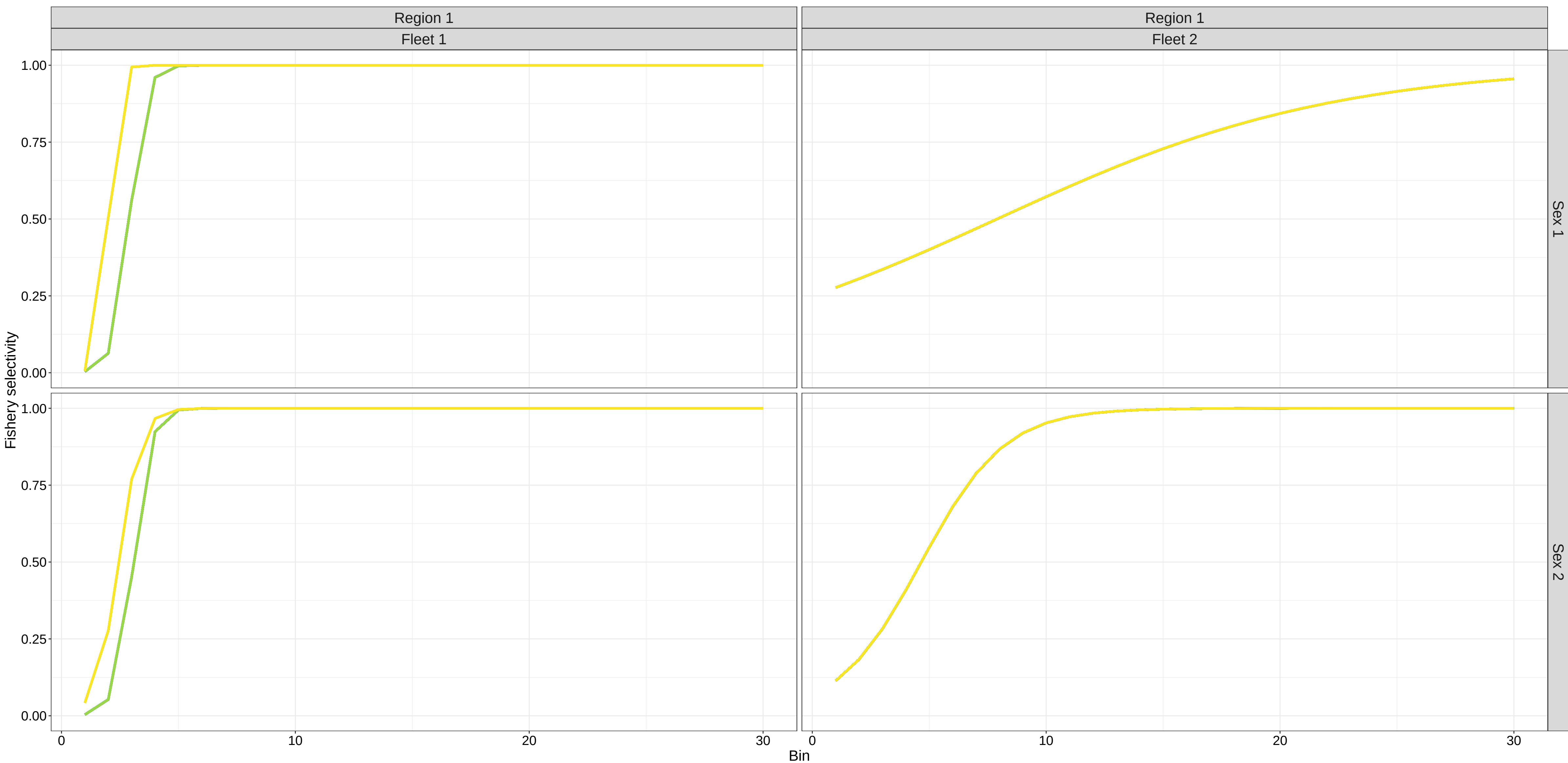
Sex — Female — Male

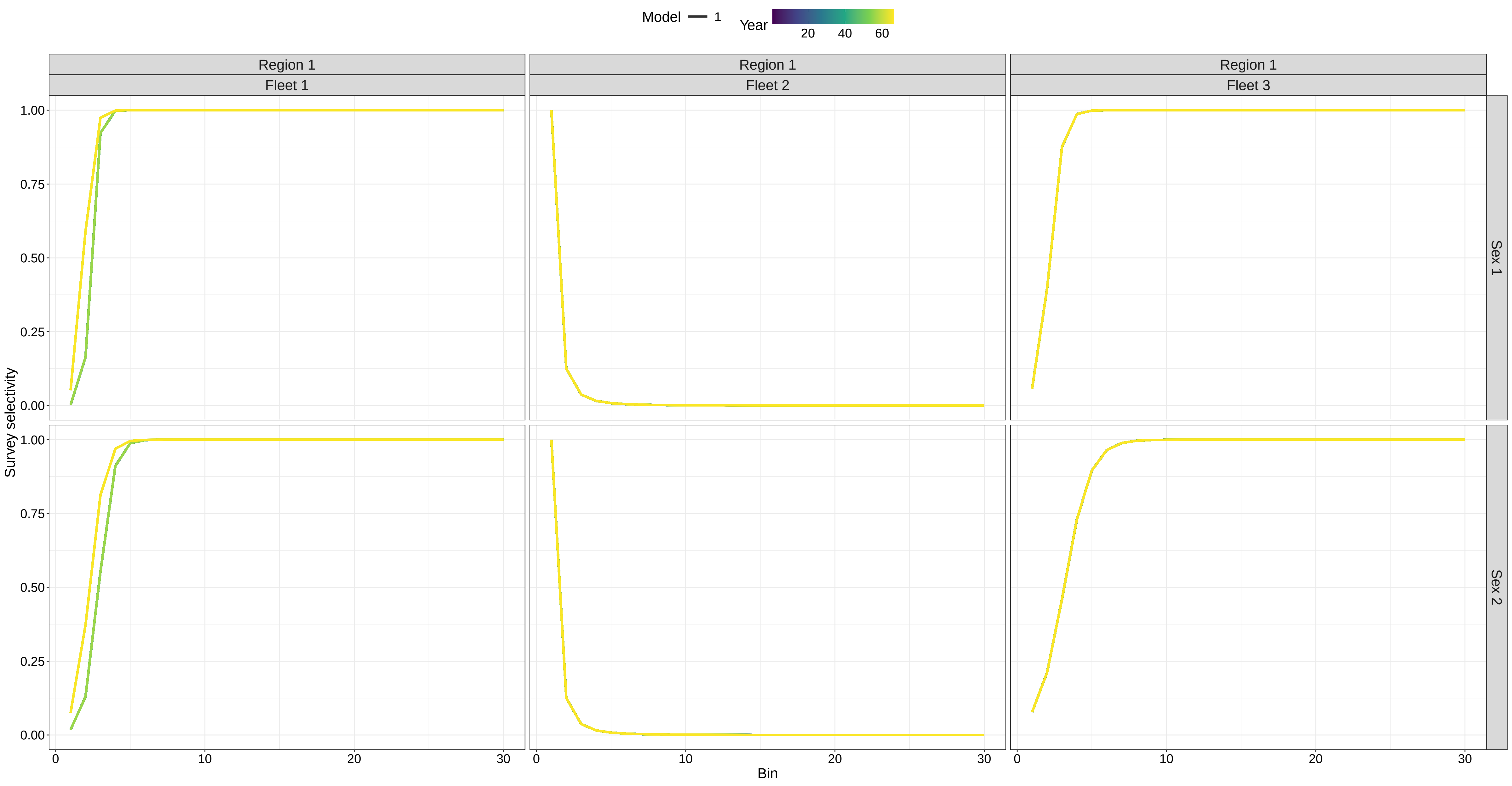
Numbers-at-age (millions)

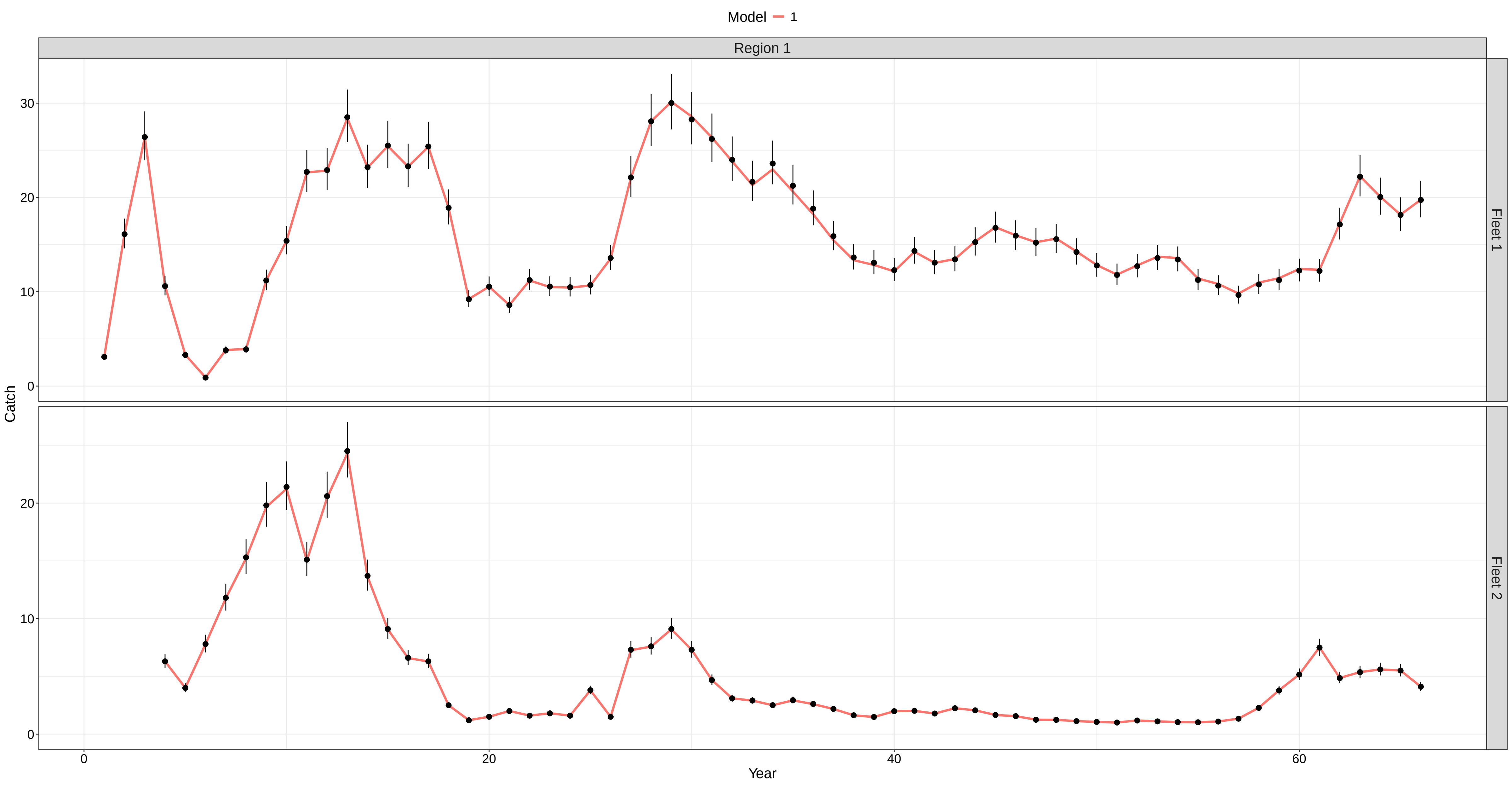


Year

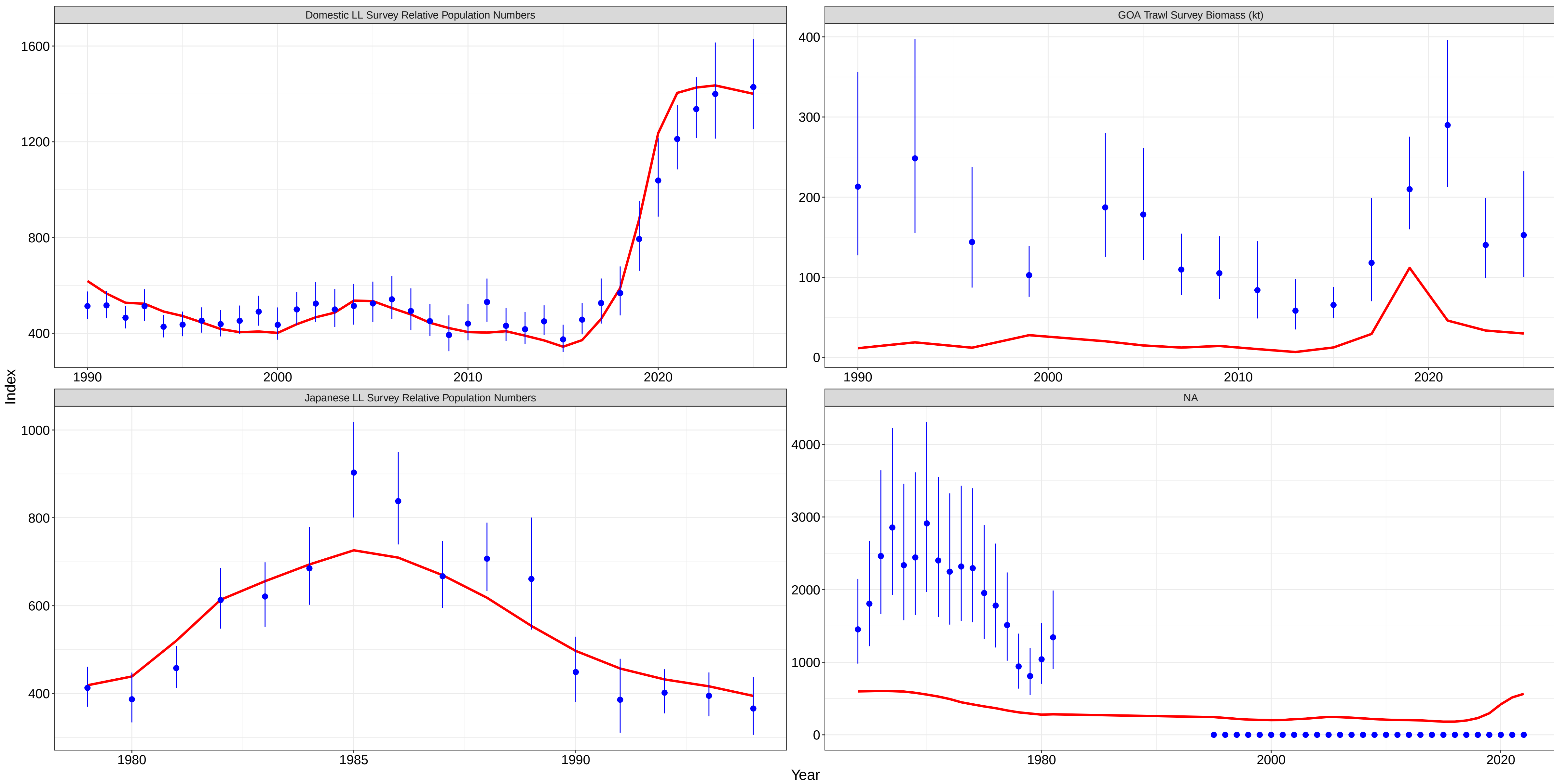
Model — 1 Year 



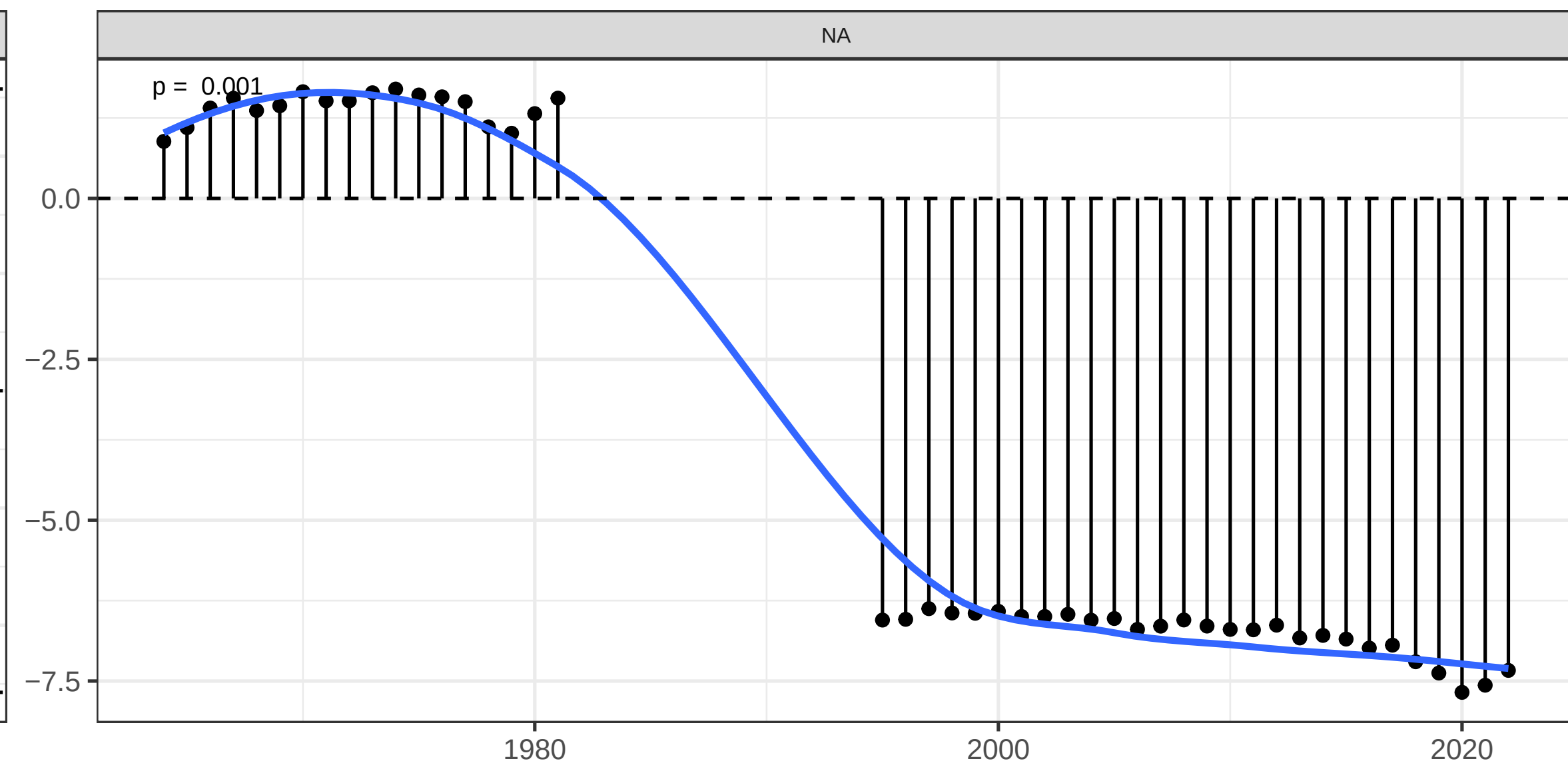
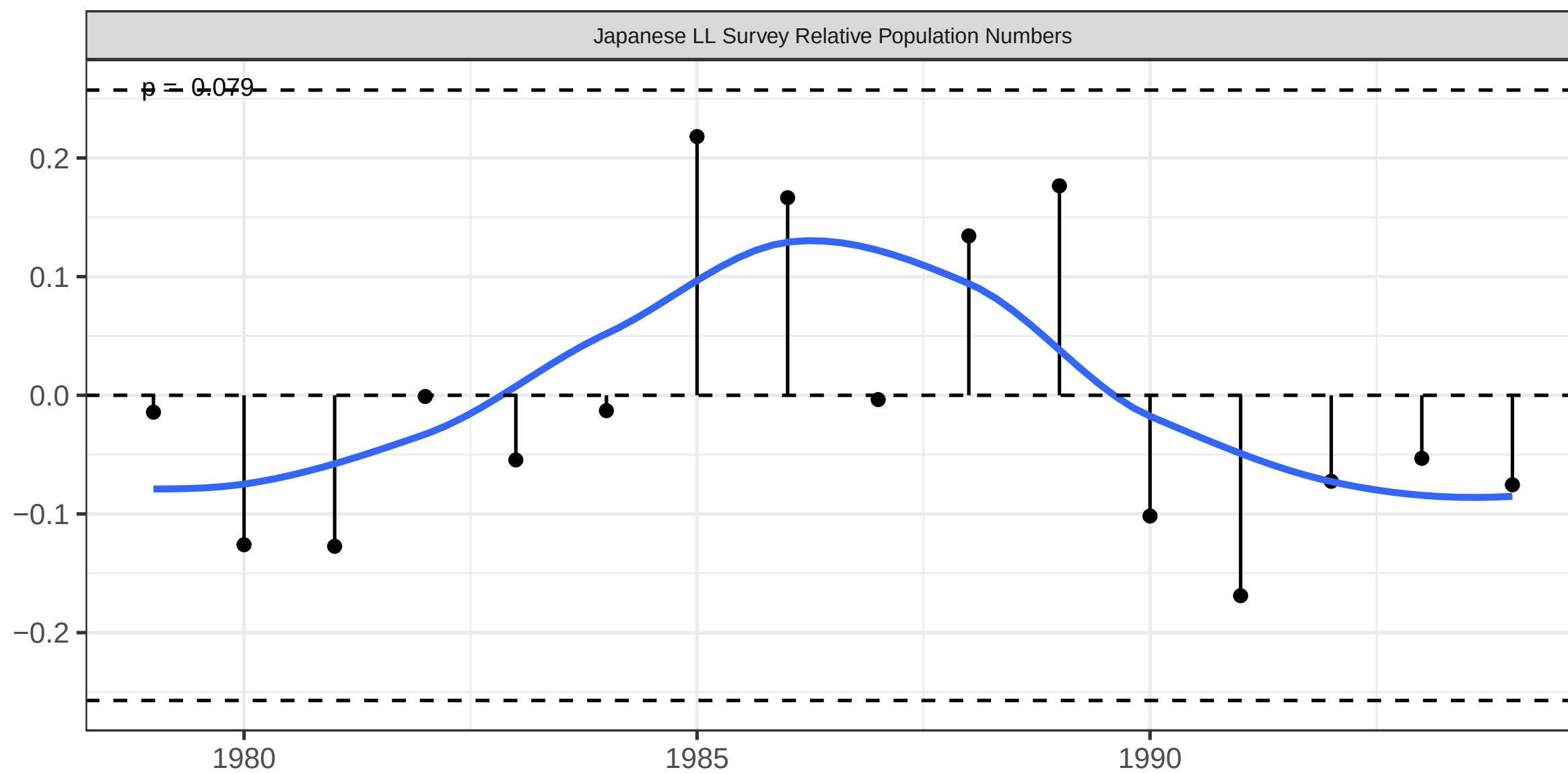
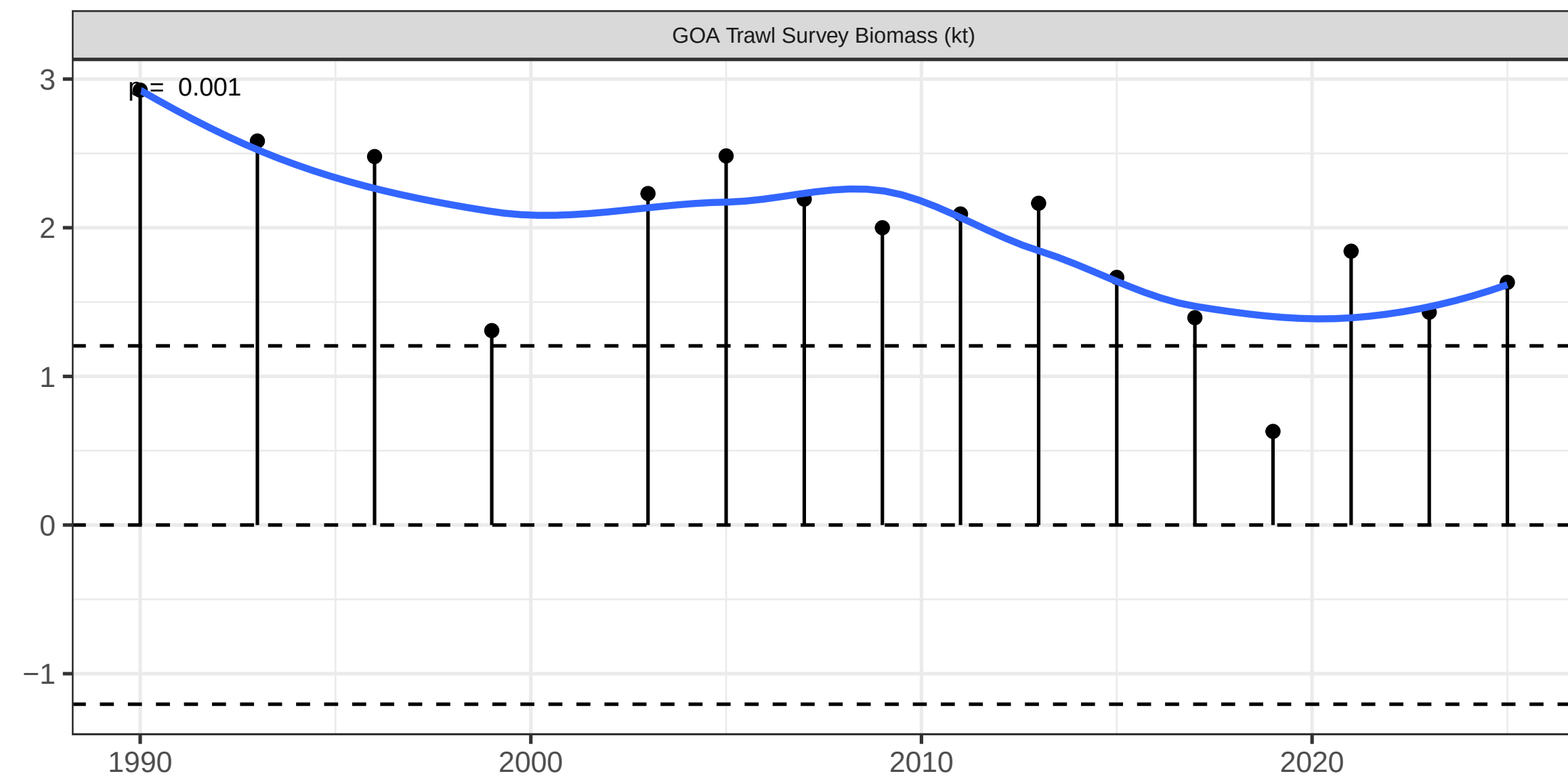
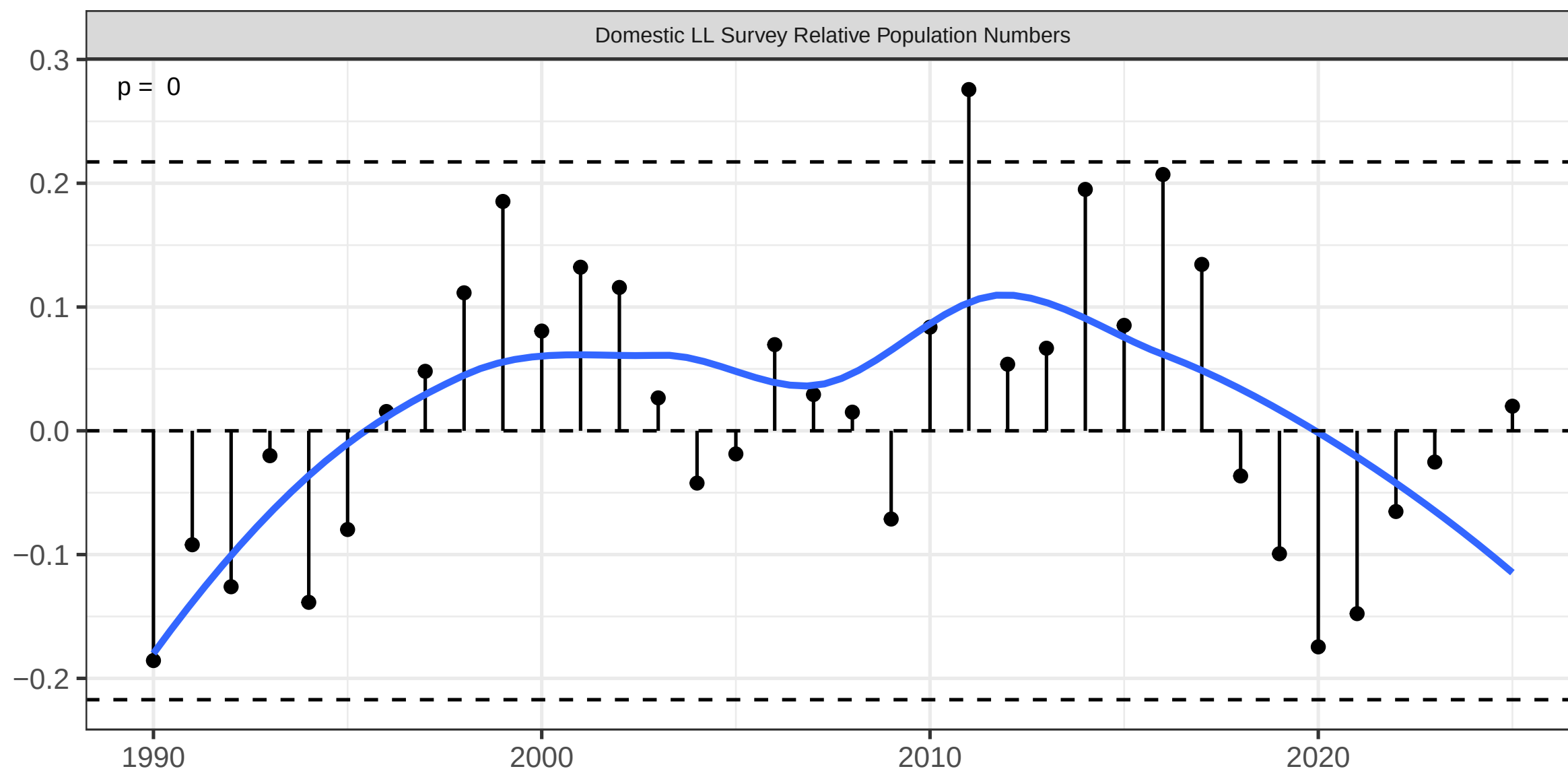




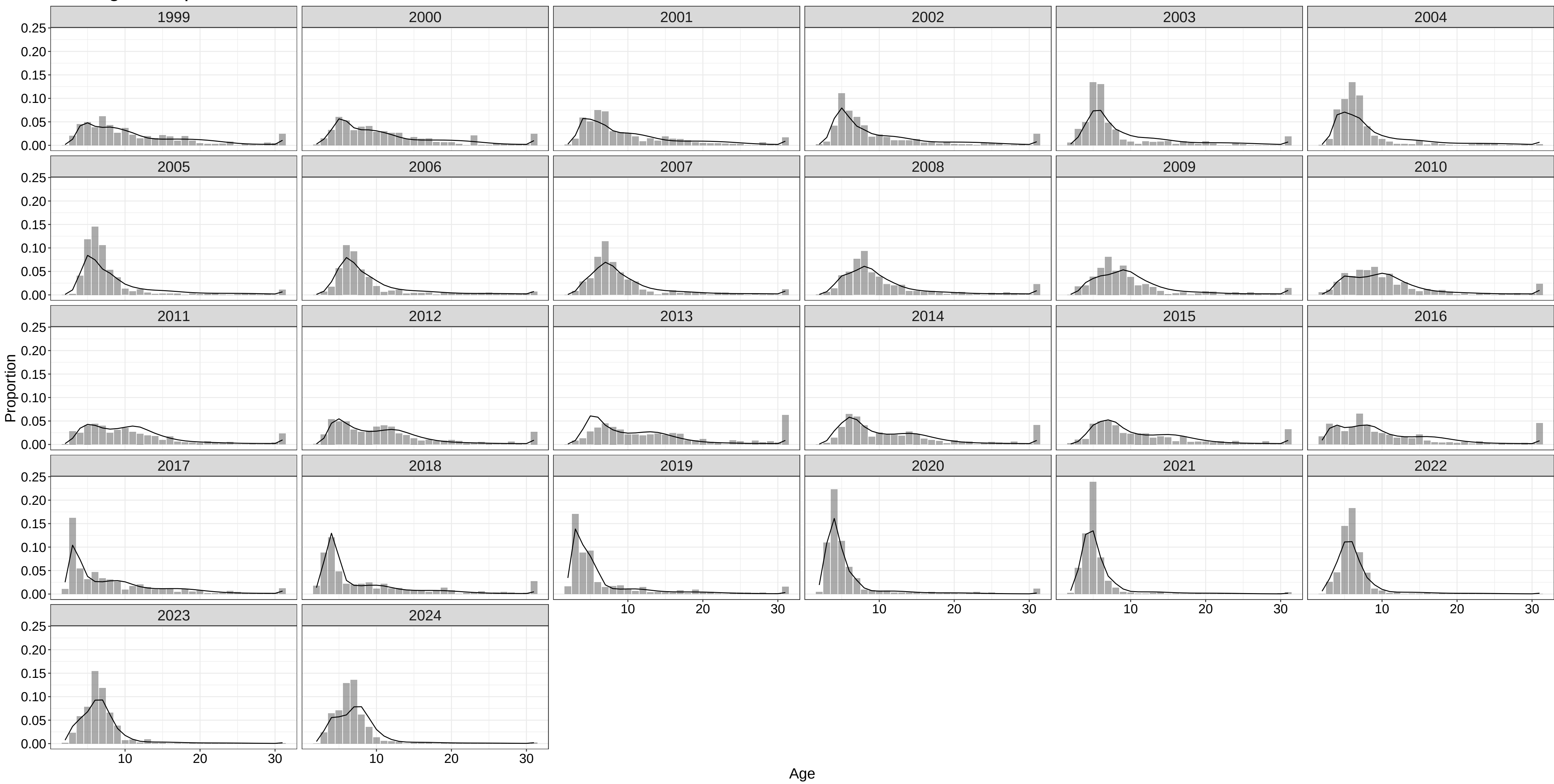
Index Fits



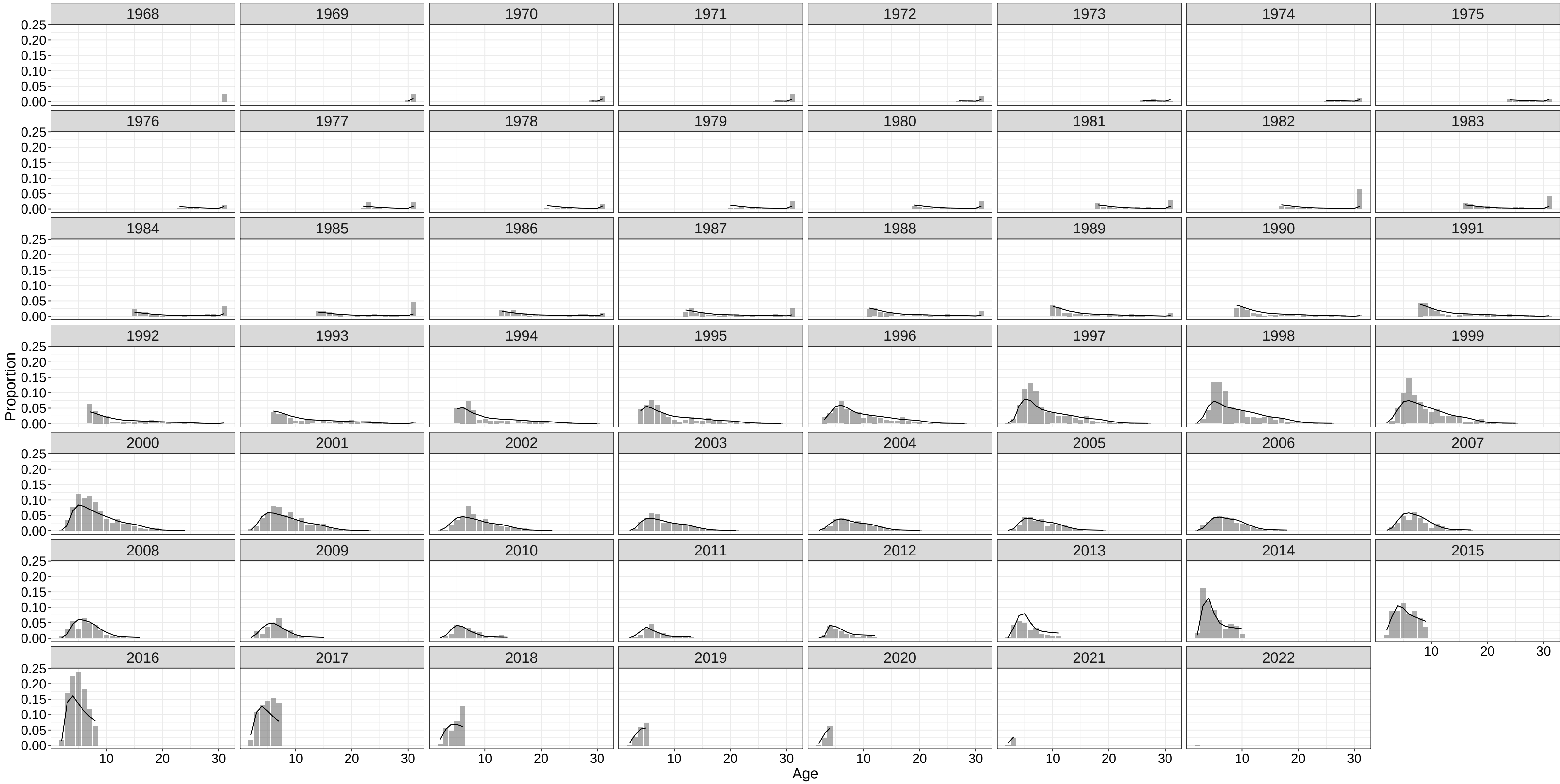
Index Residuals Runs Test



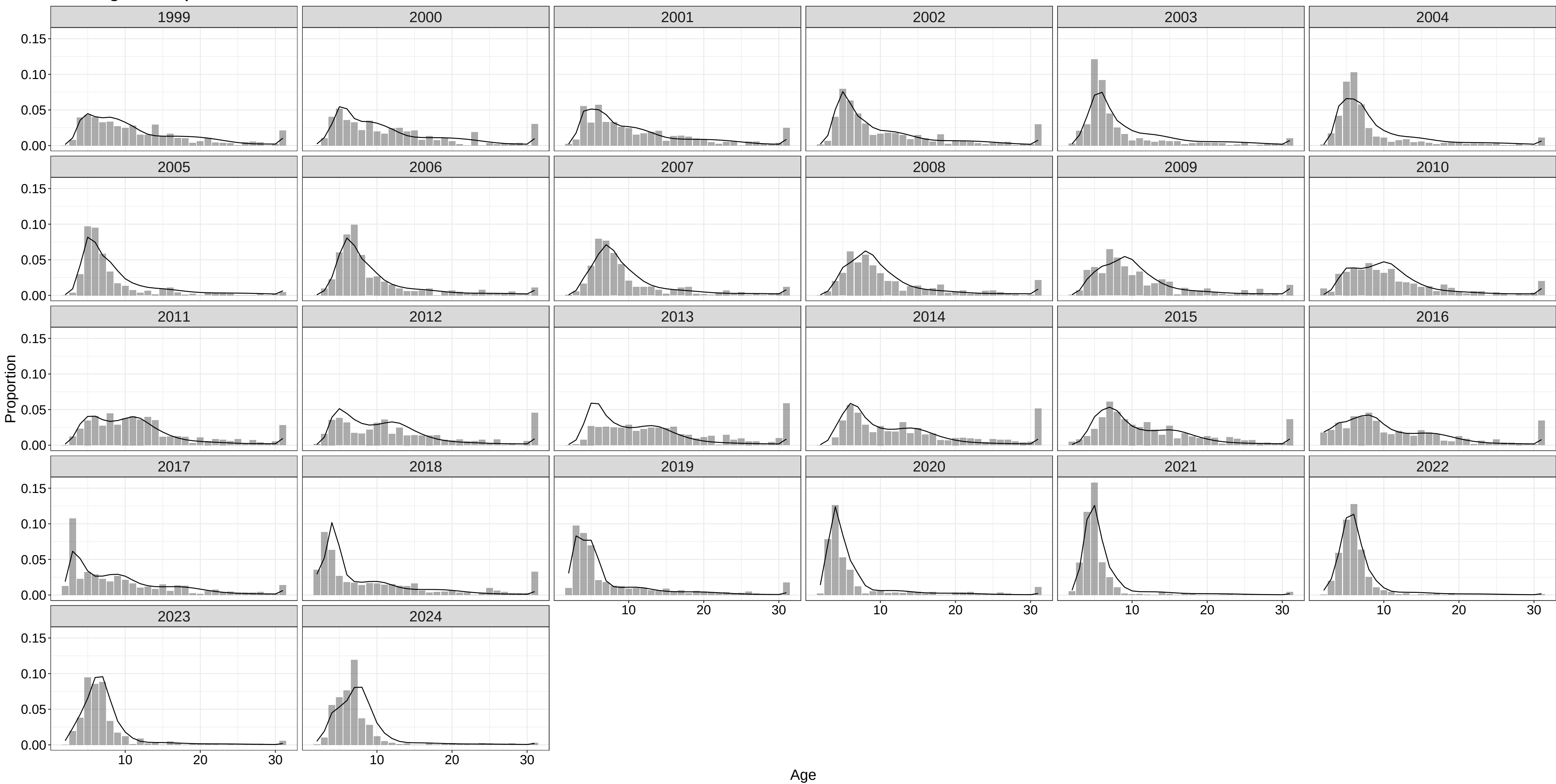
LLF Age Comps Female



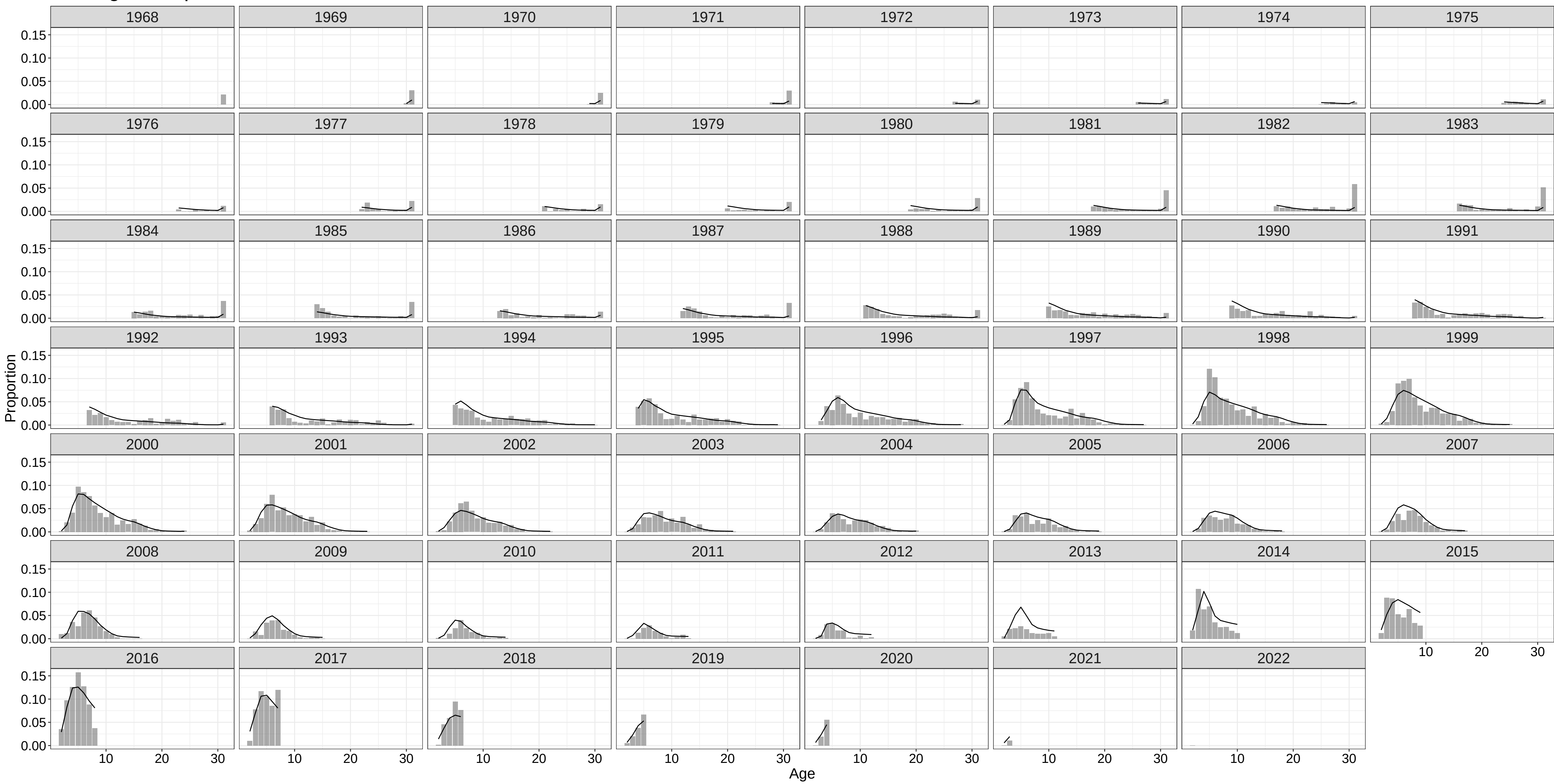
LLF Age Comps Cohort Fits Female



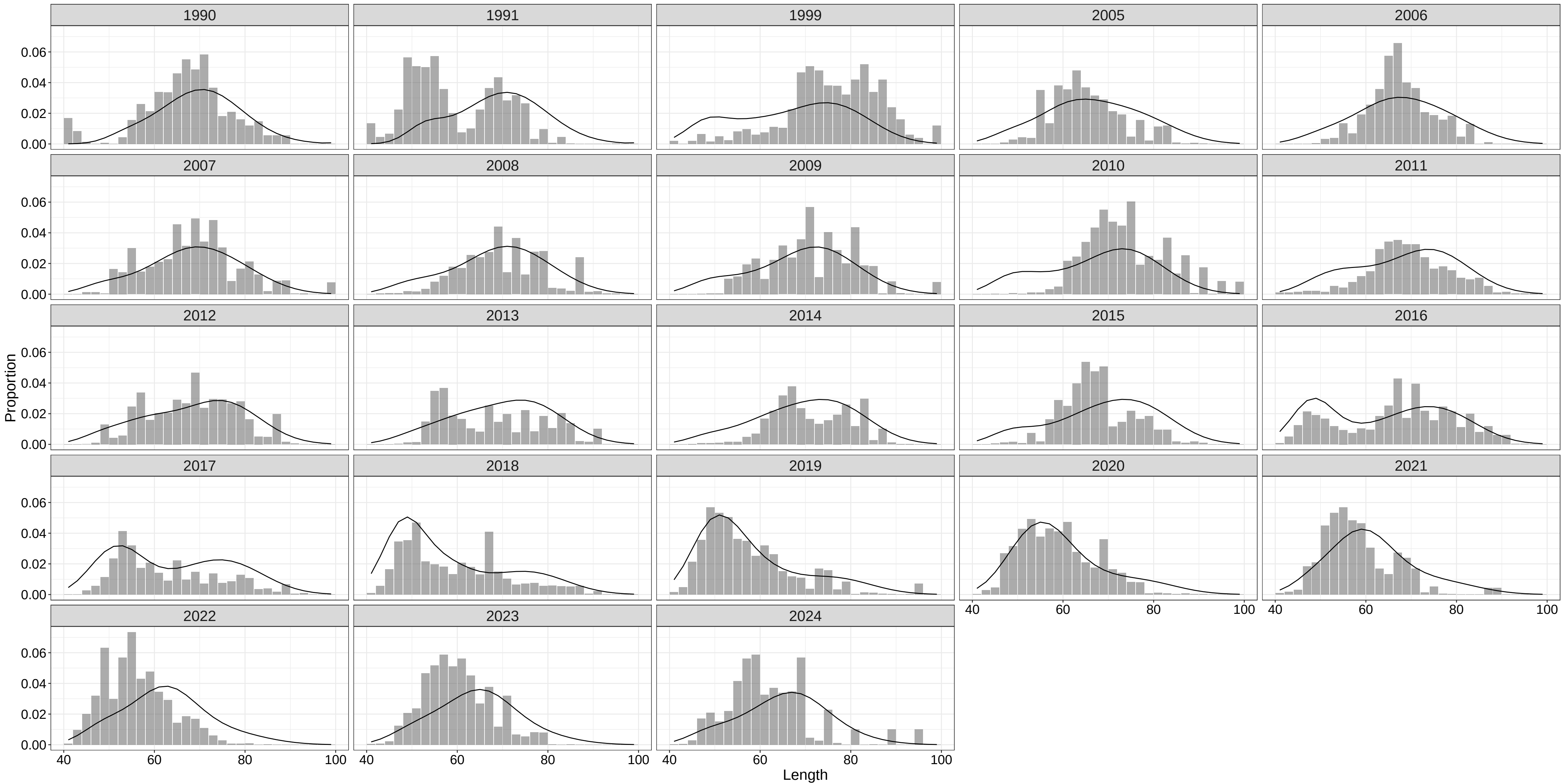
LLF Age Comps Male



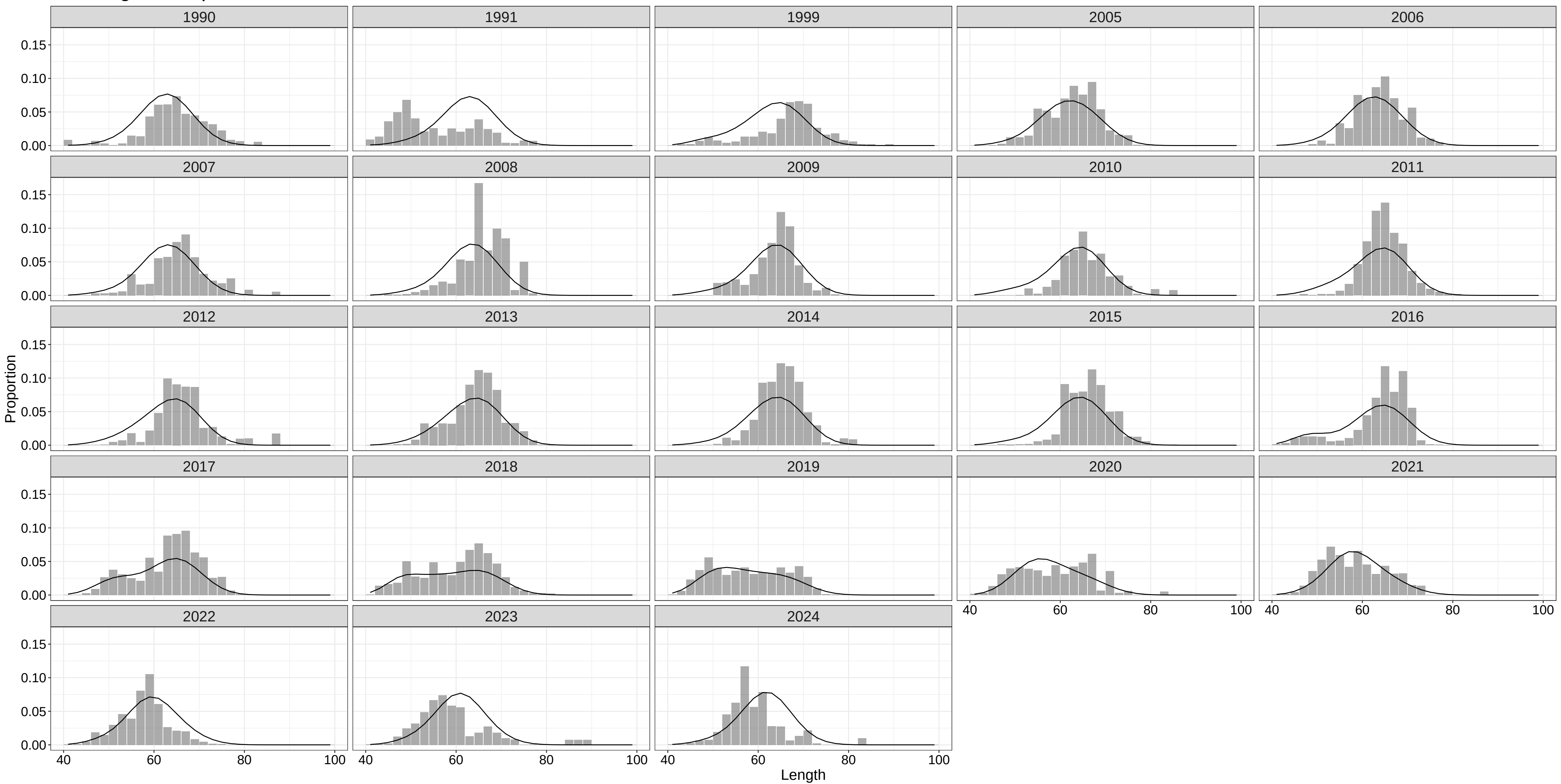
LLF Age Comps Cohort Fits Male



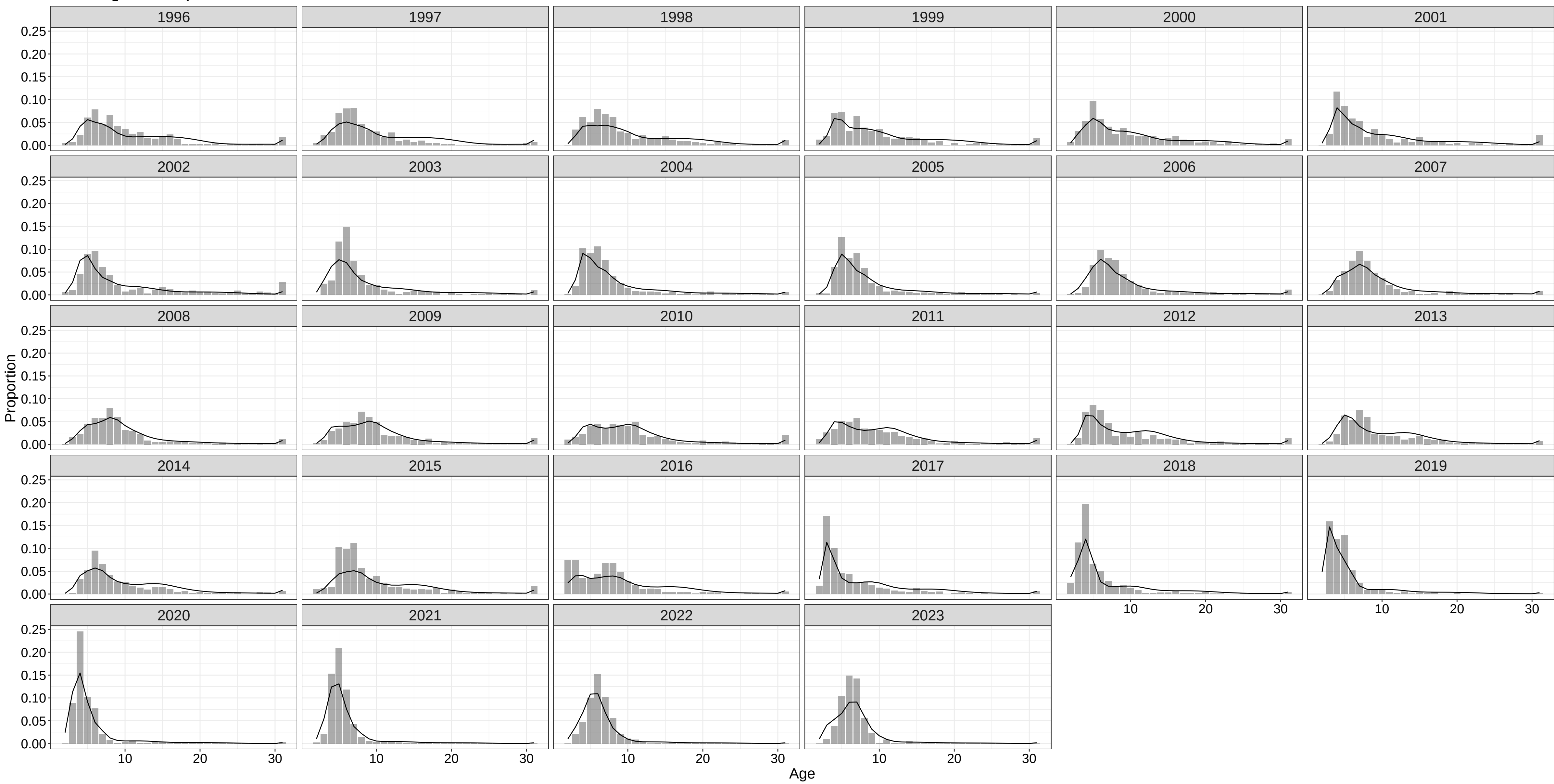
TF Length Comps Female



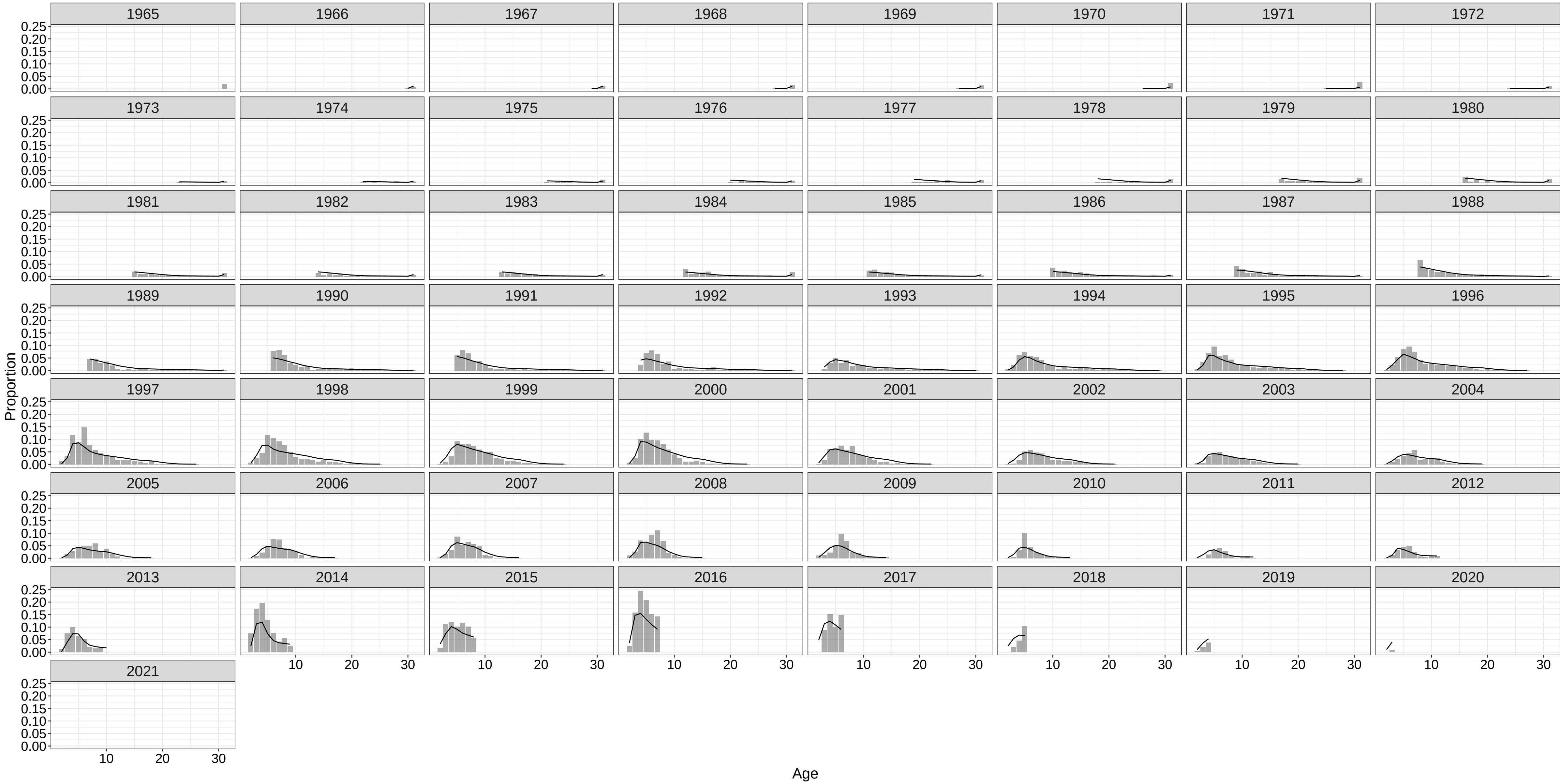
TF Length Comps Male



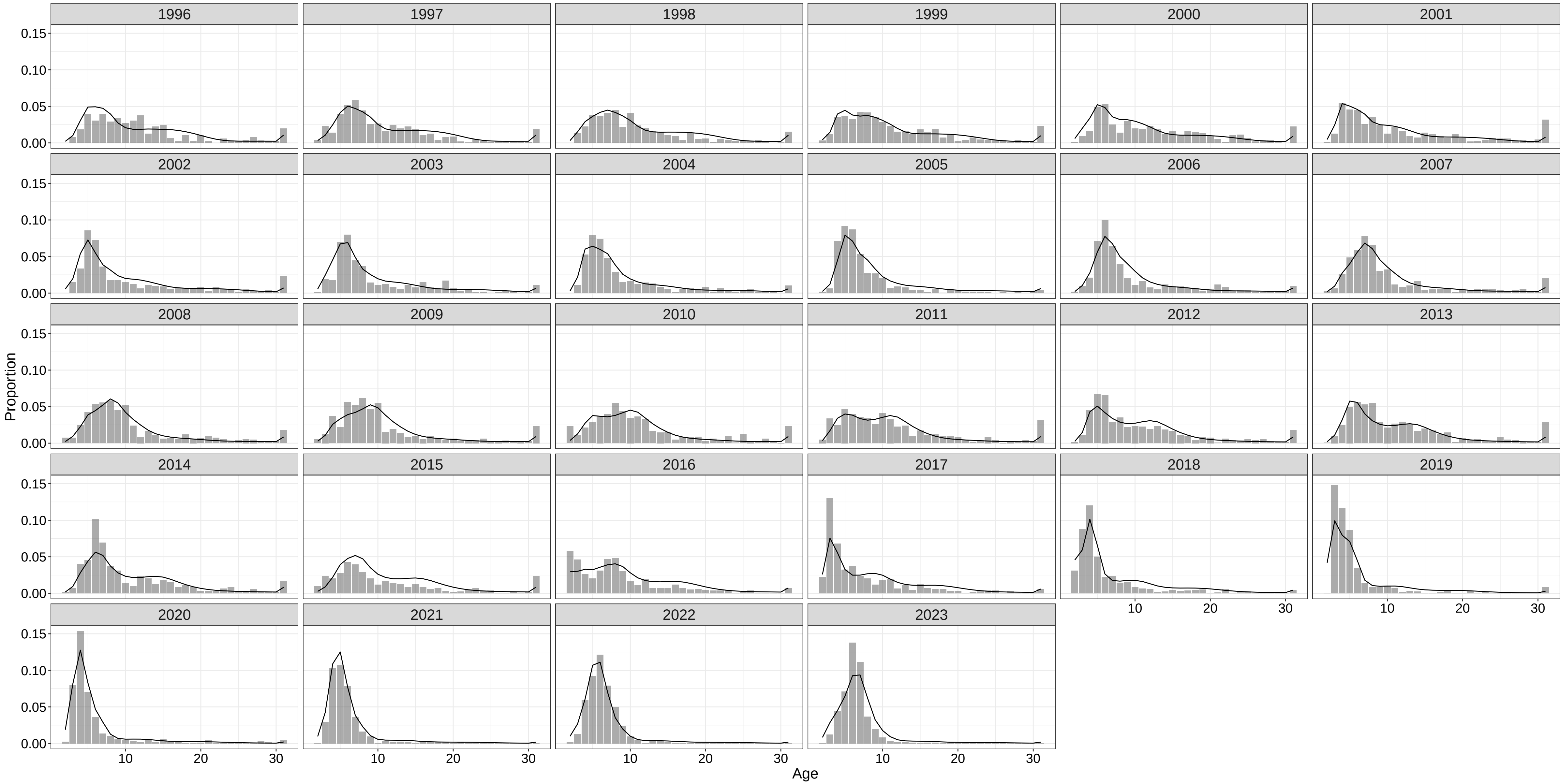
LLS Age Comps Female



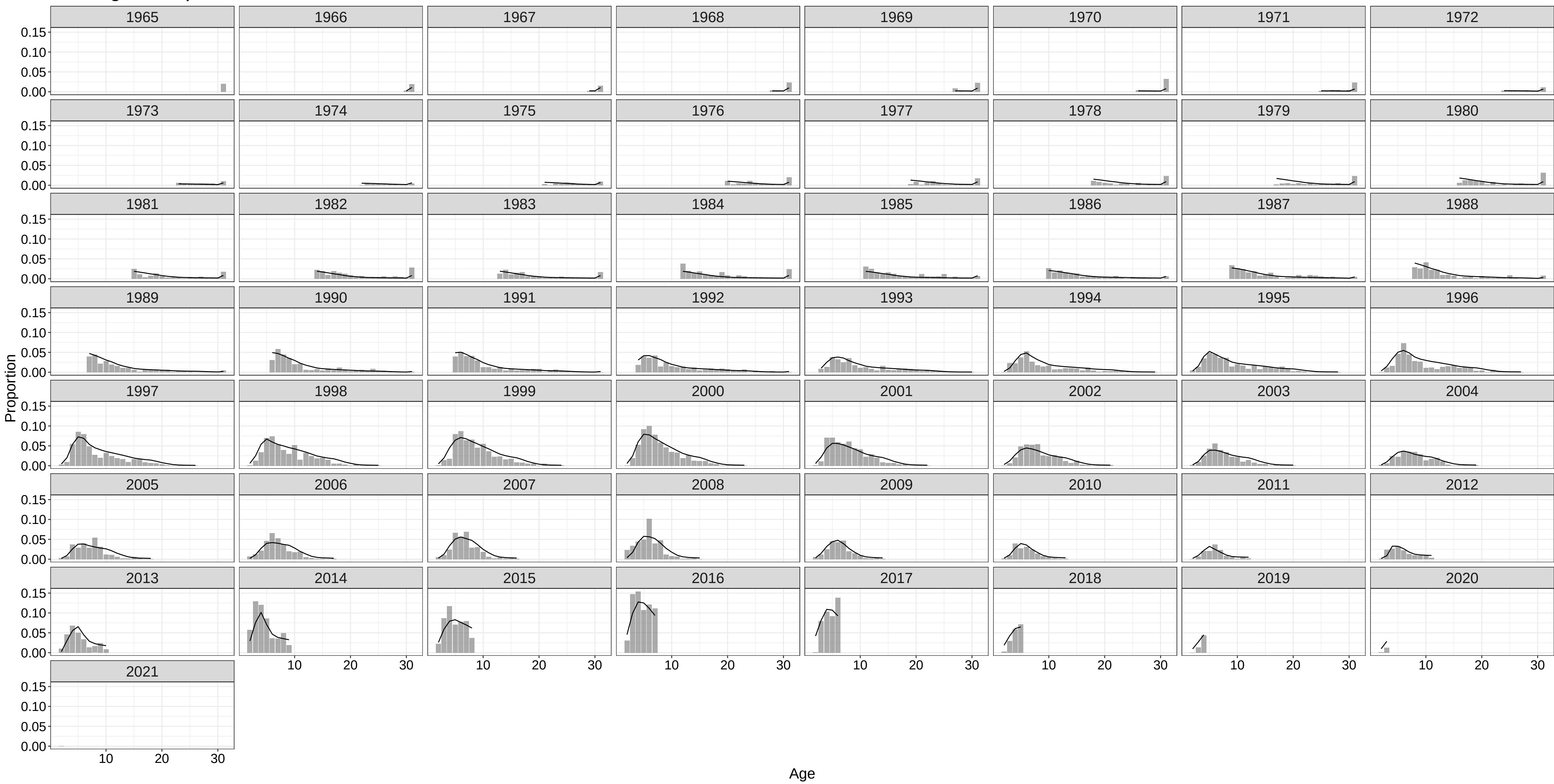
LLS Age Comps Cohort Fit Female



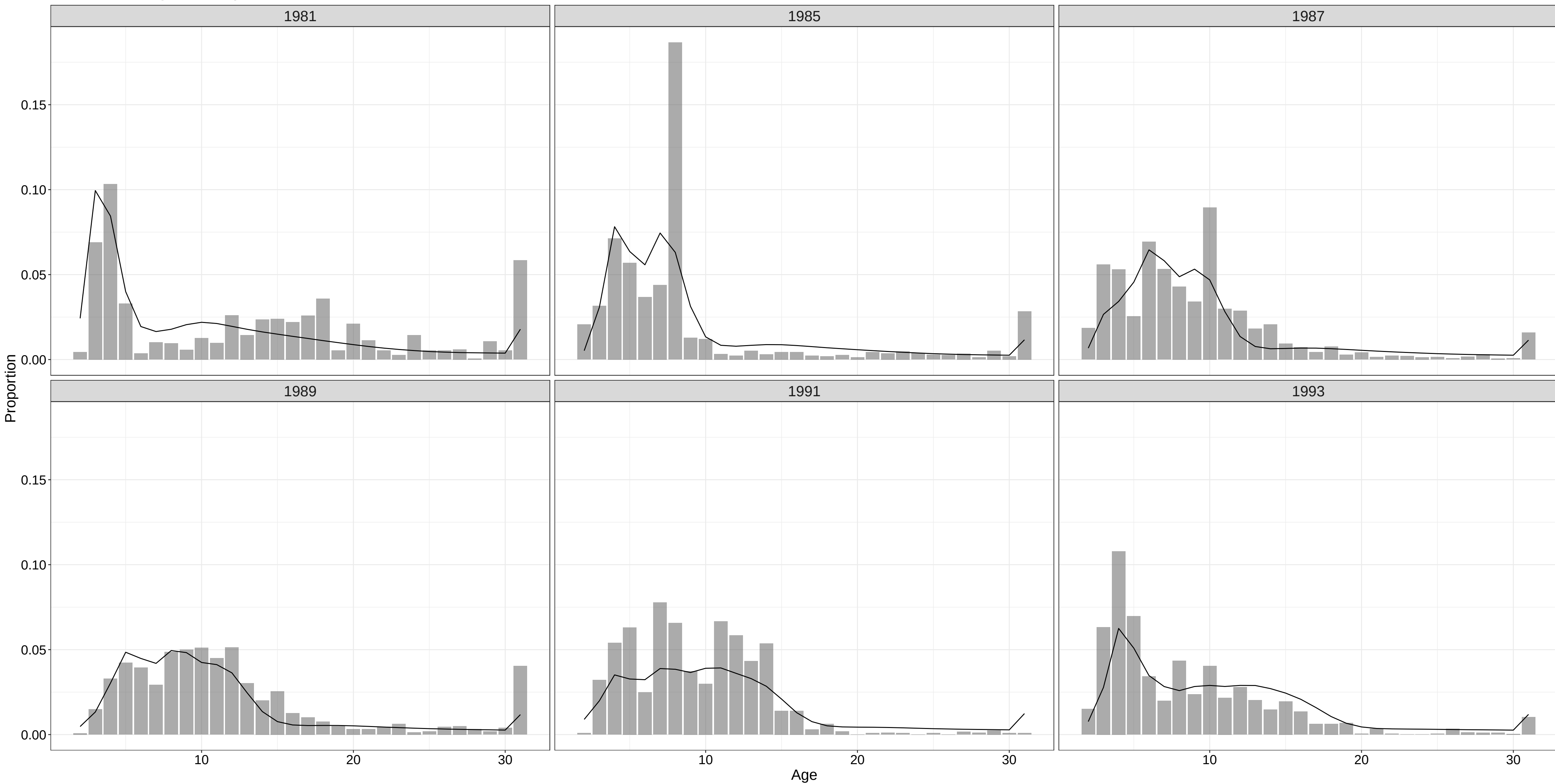
LLS Age Comps Male



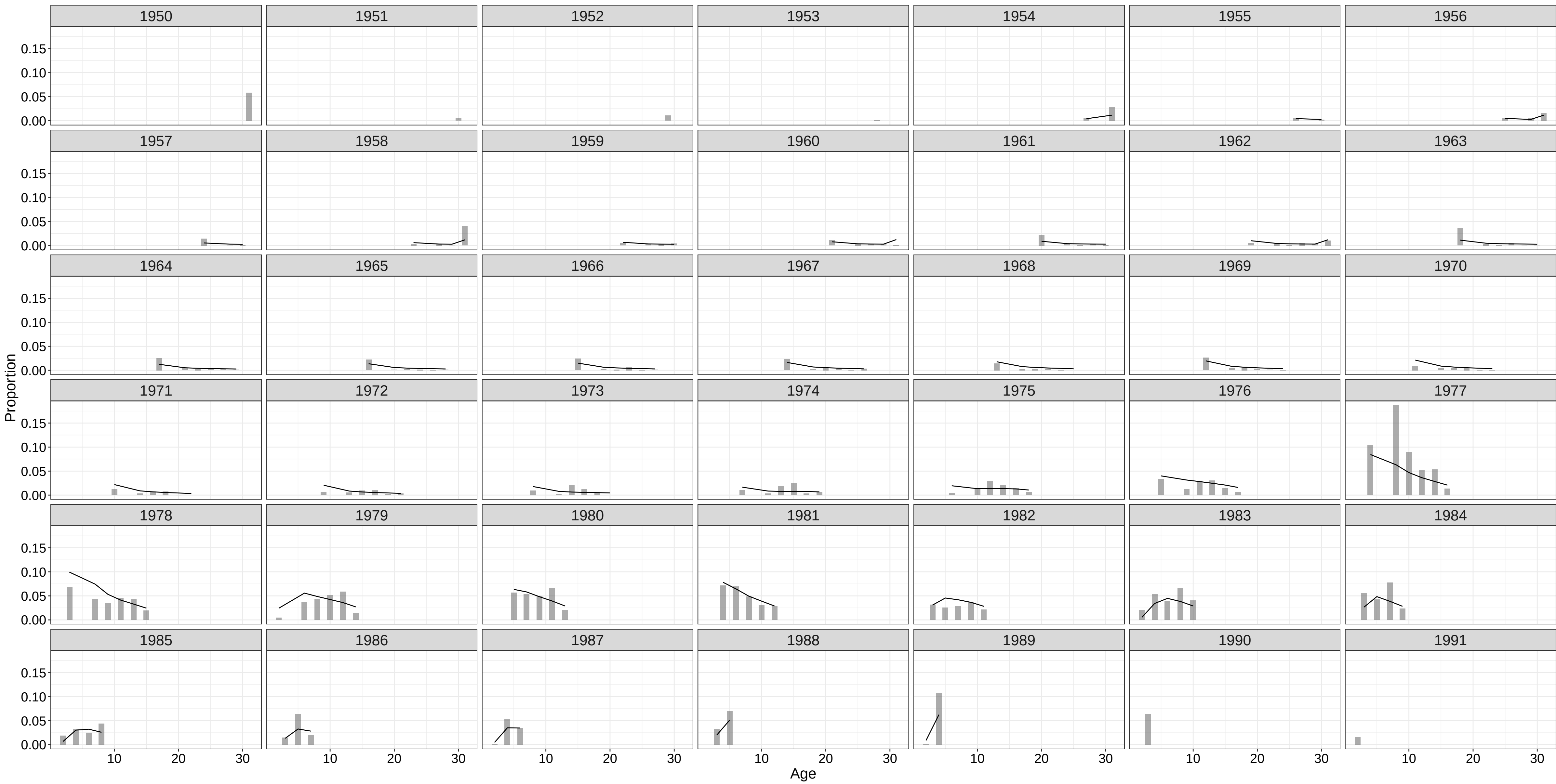
LLS Age Comps Cohort Fit Male



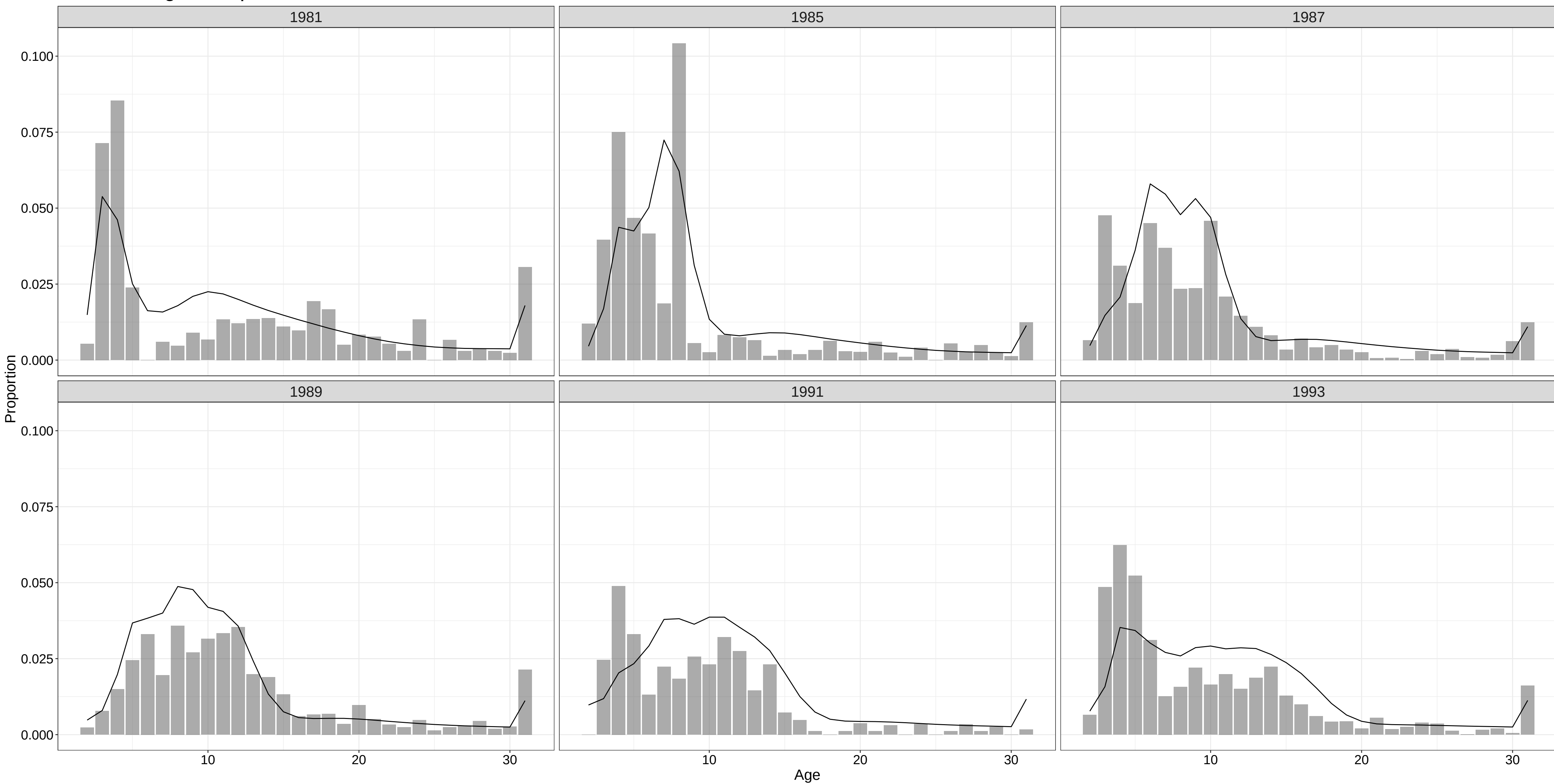
JPN LLS Age Comps Female



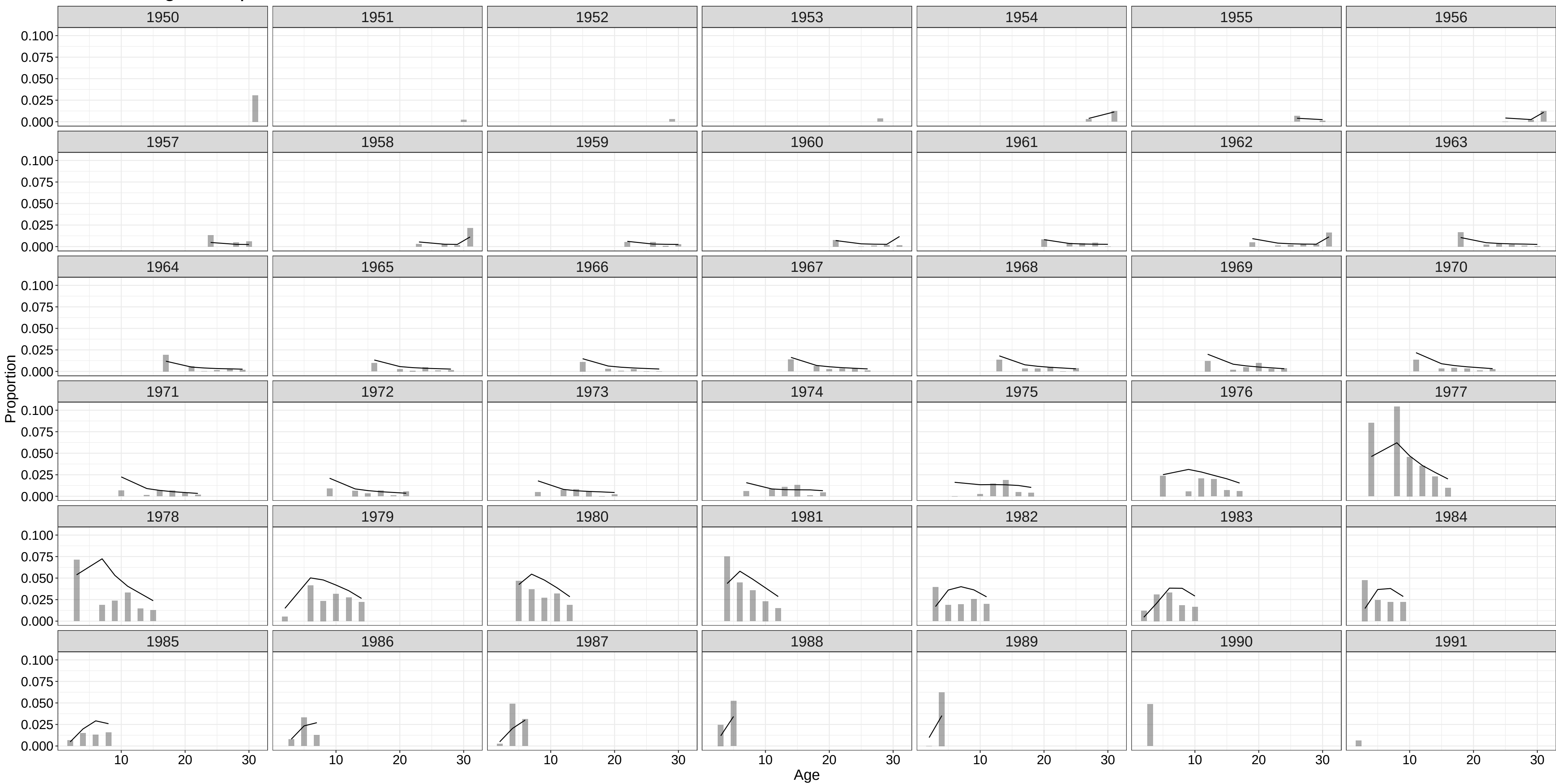
JPN LLS Age Comps Cohort Fit Female



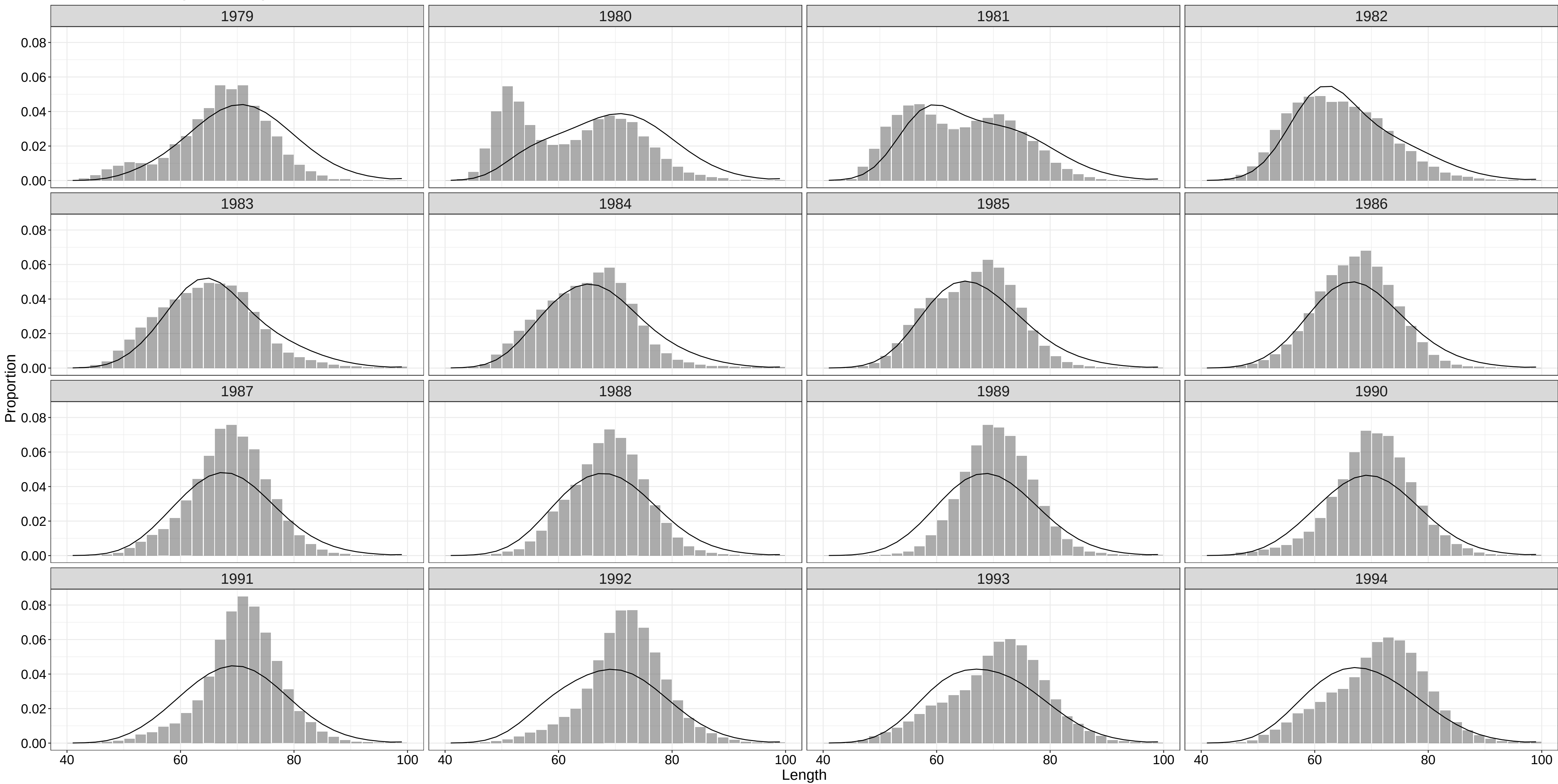
JPN LLS Age Comps Male



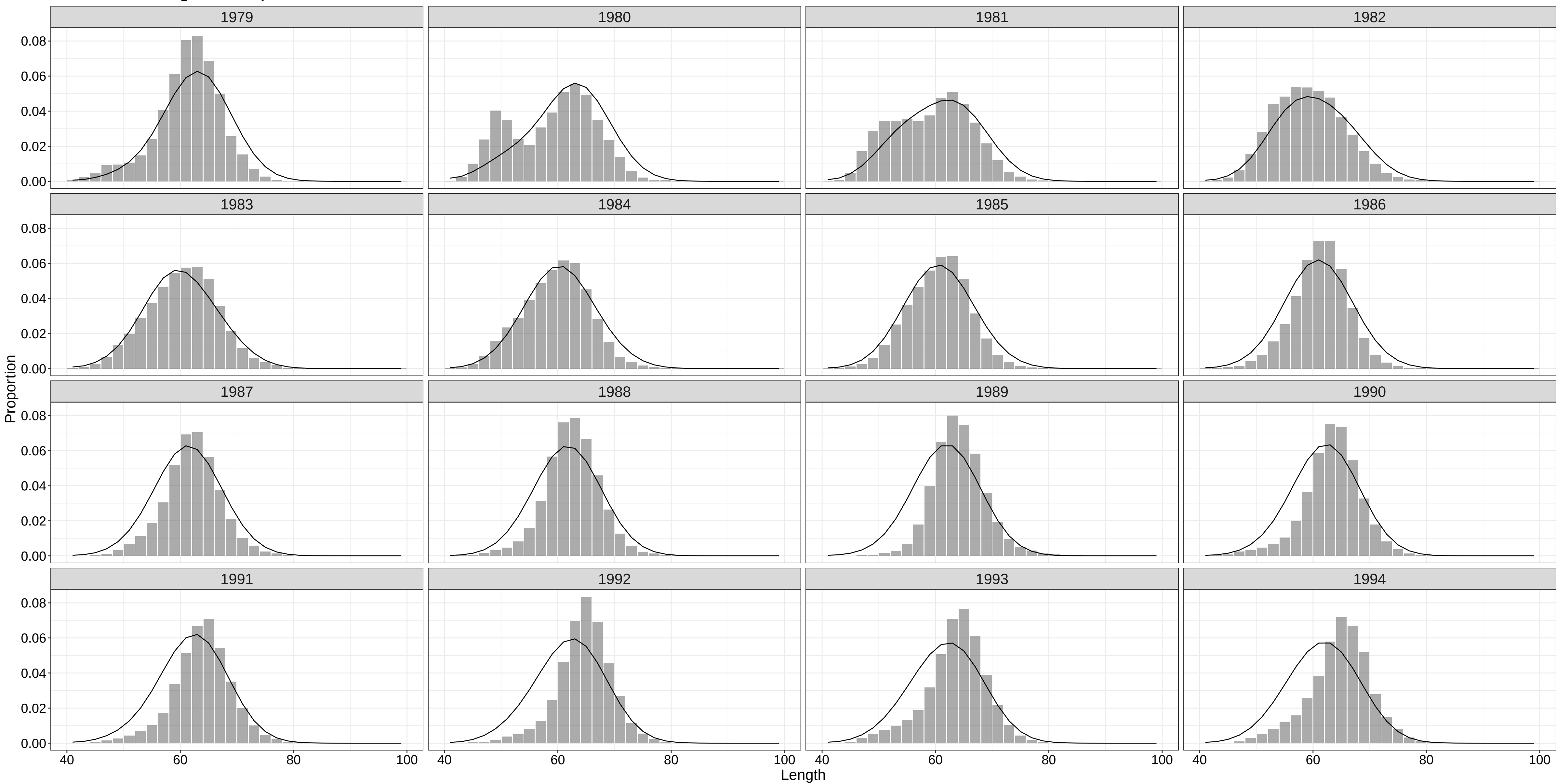
JPN LLS Age Comps Cohort Fit Male

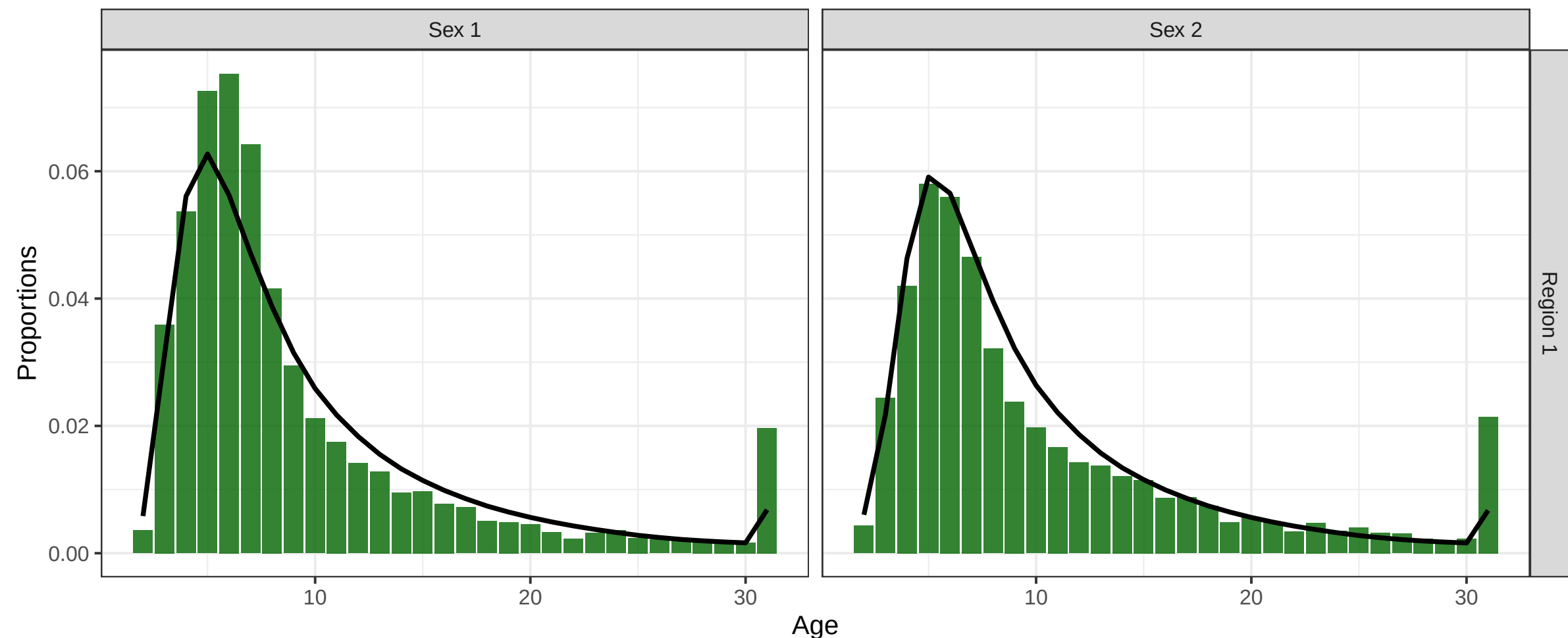
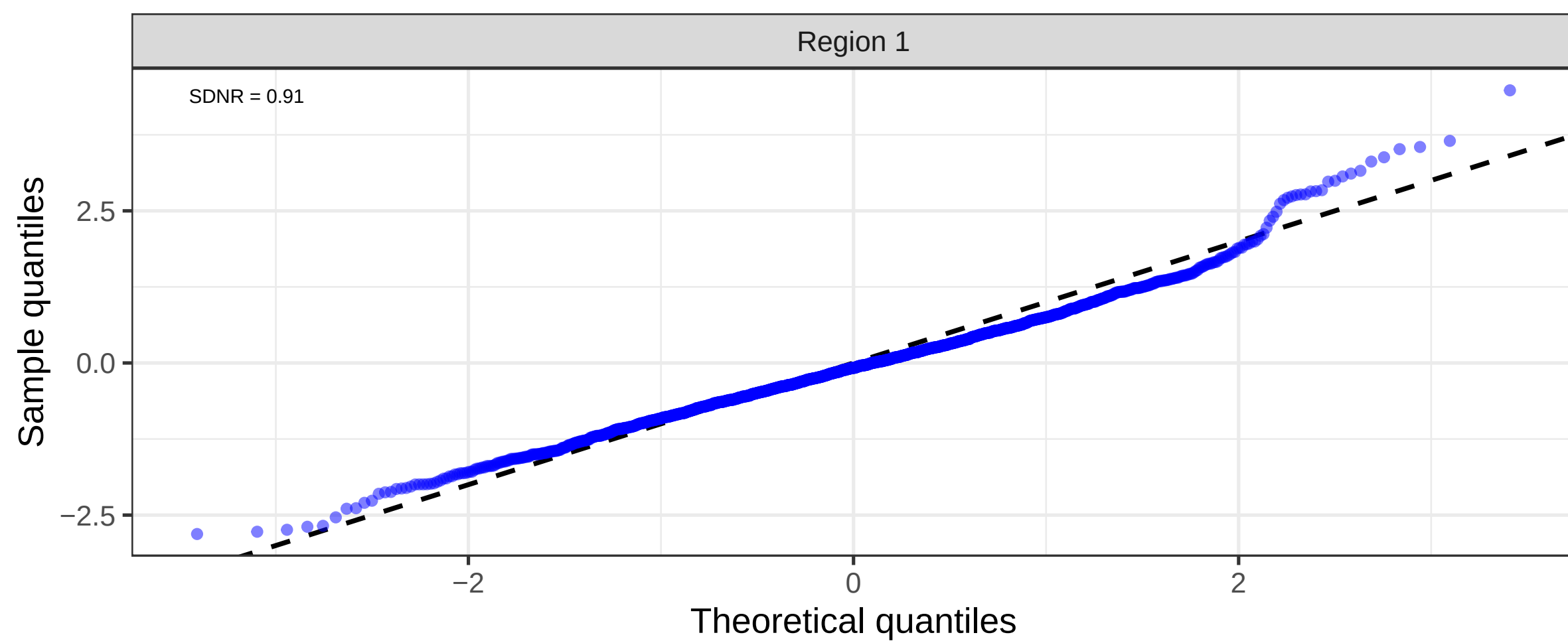
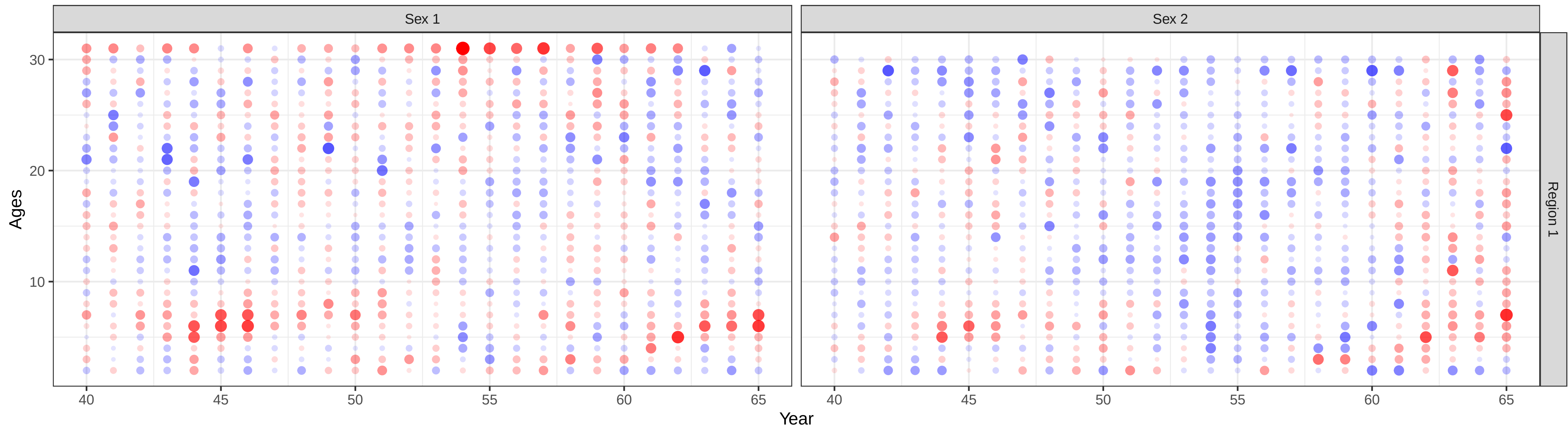


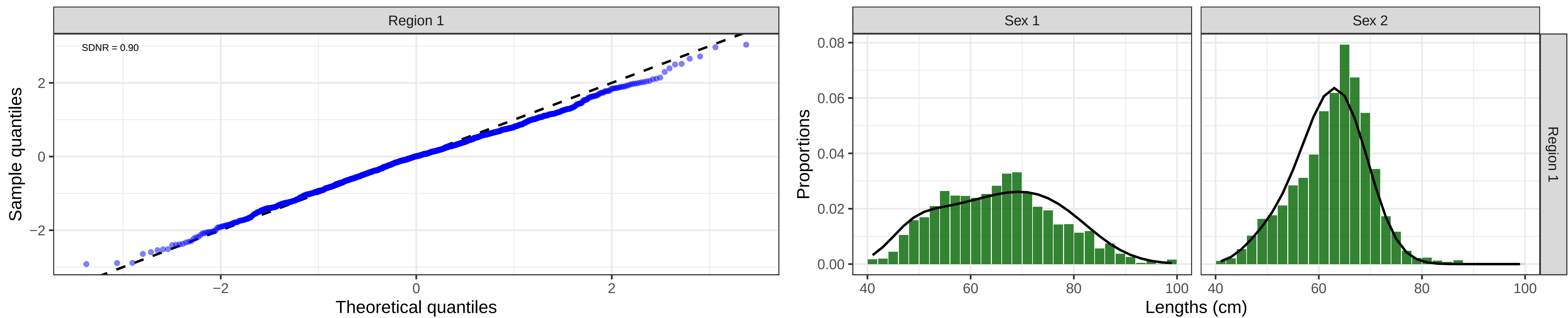
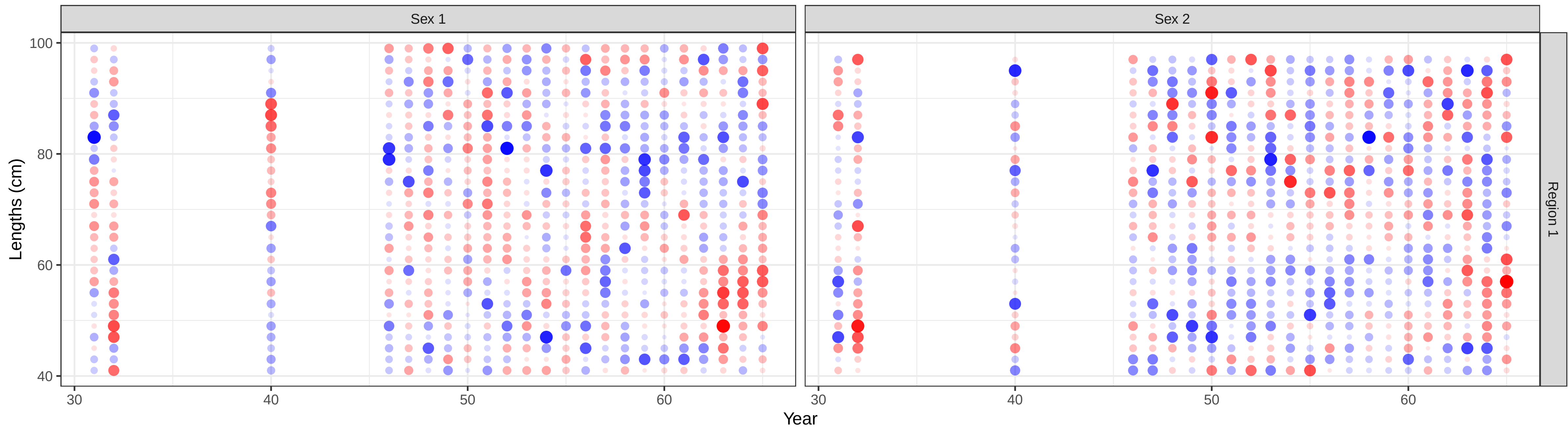
JPN LLS Length Comps Female



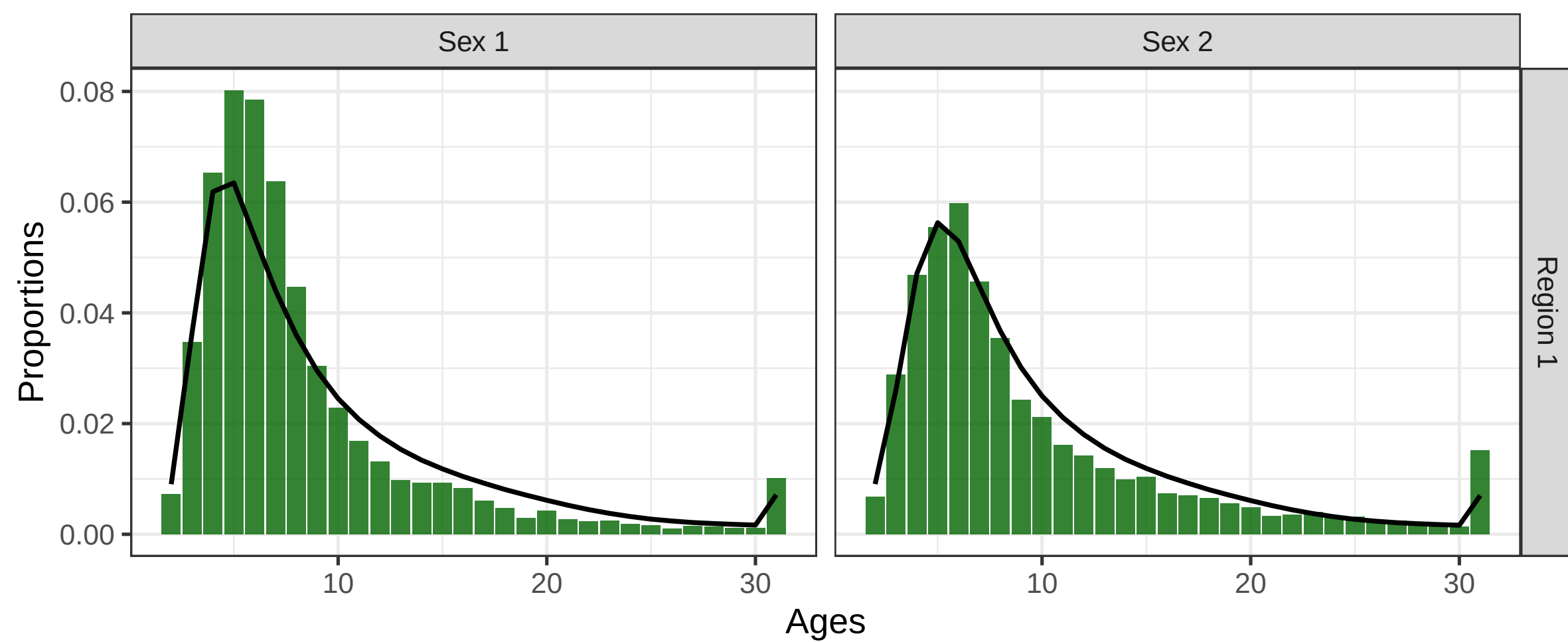
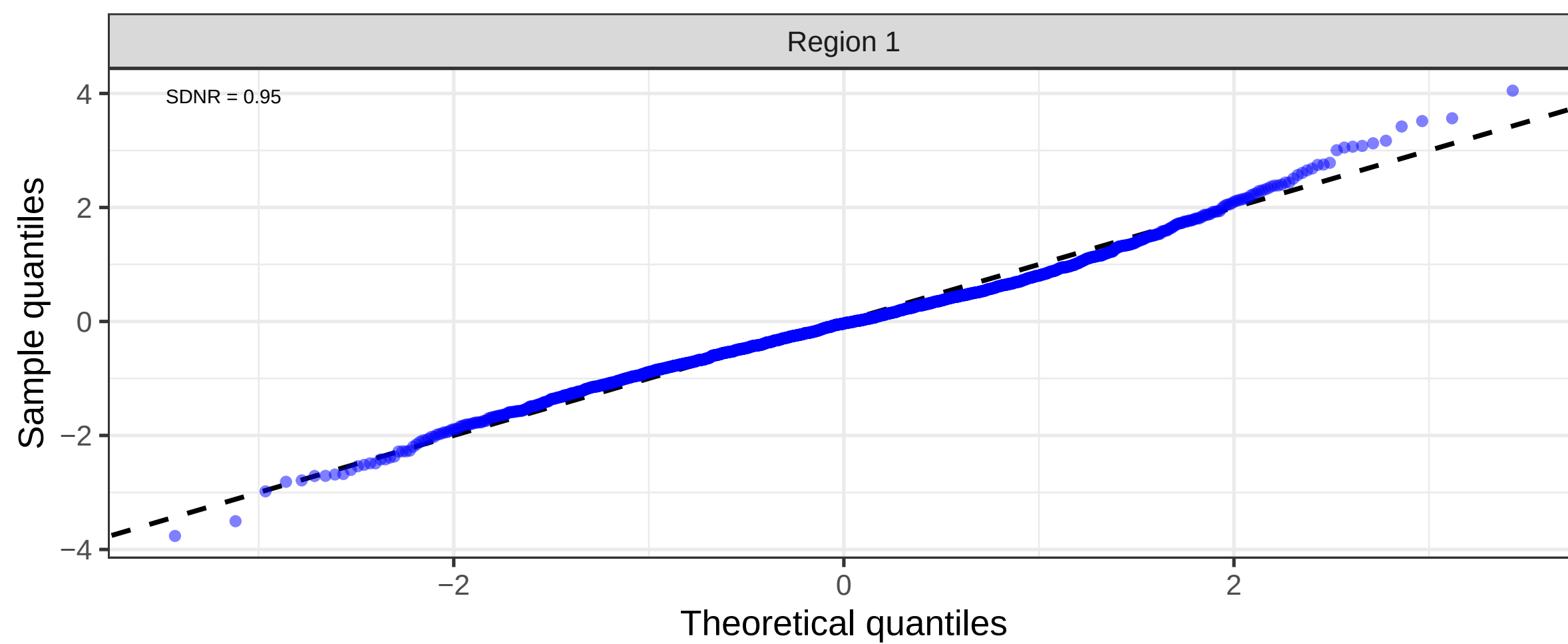
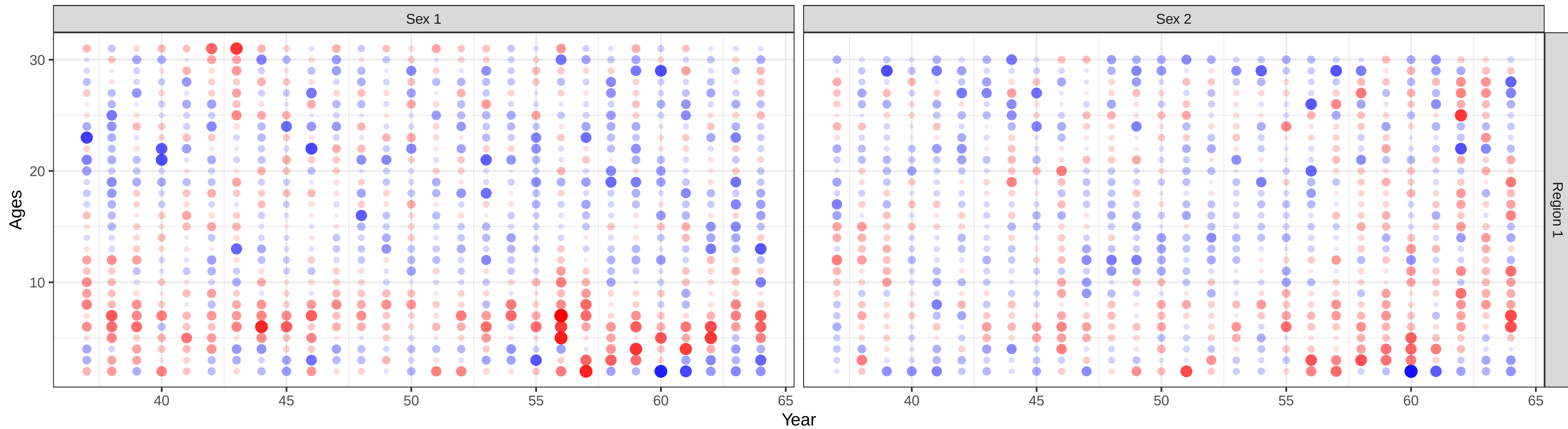
JPN LLS Length Comps Male



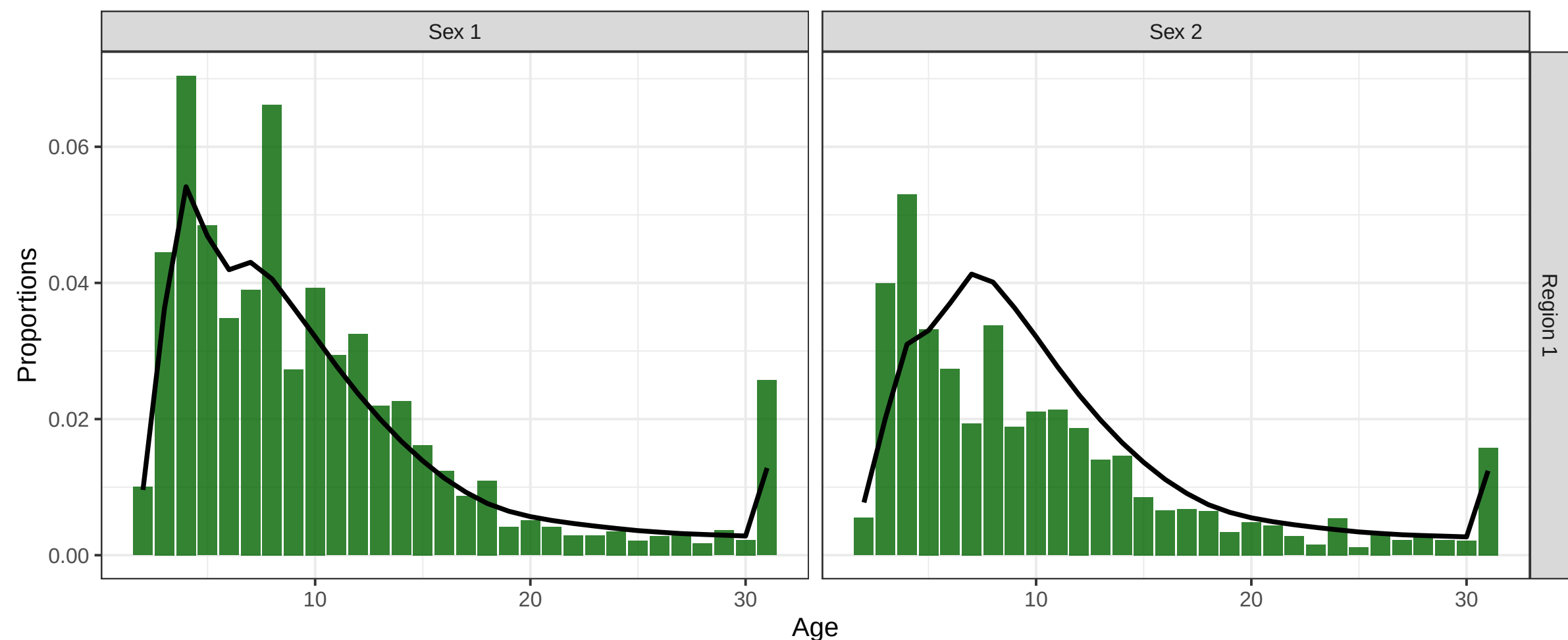
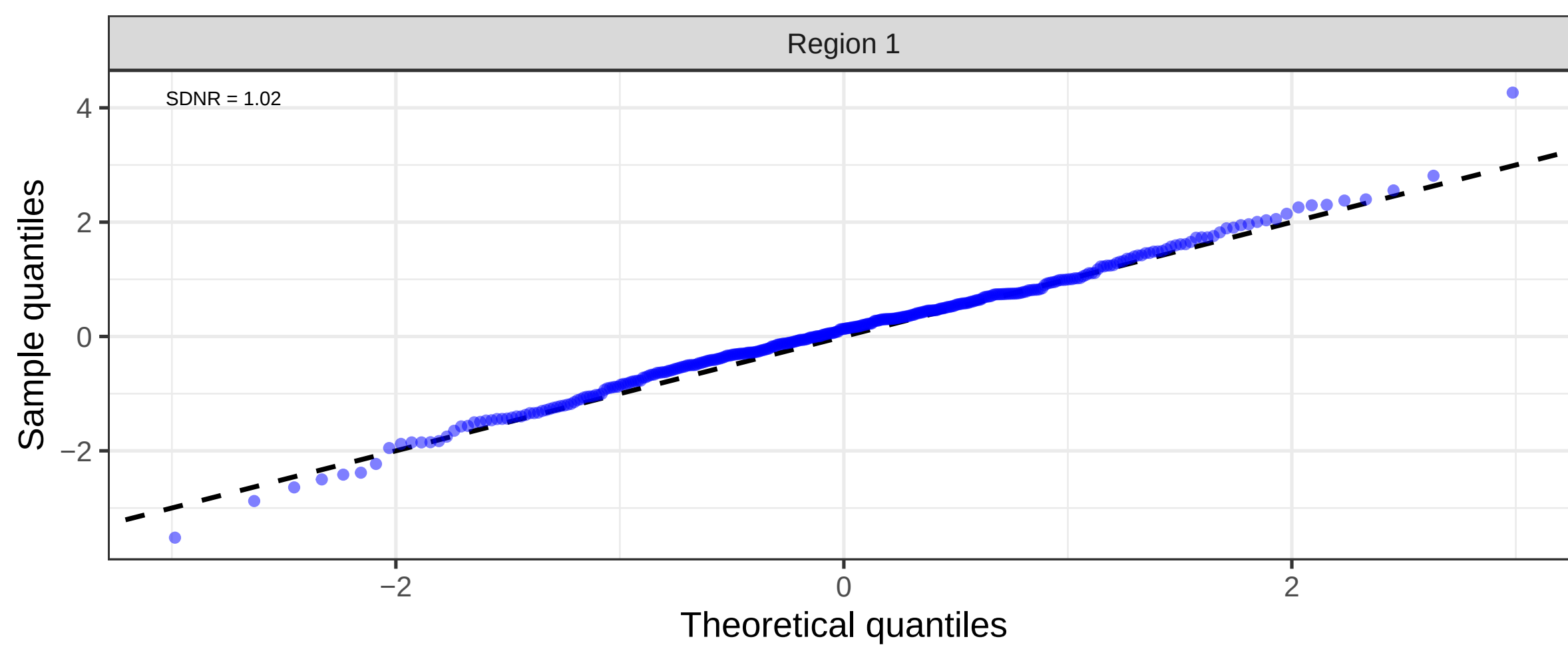
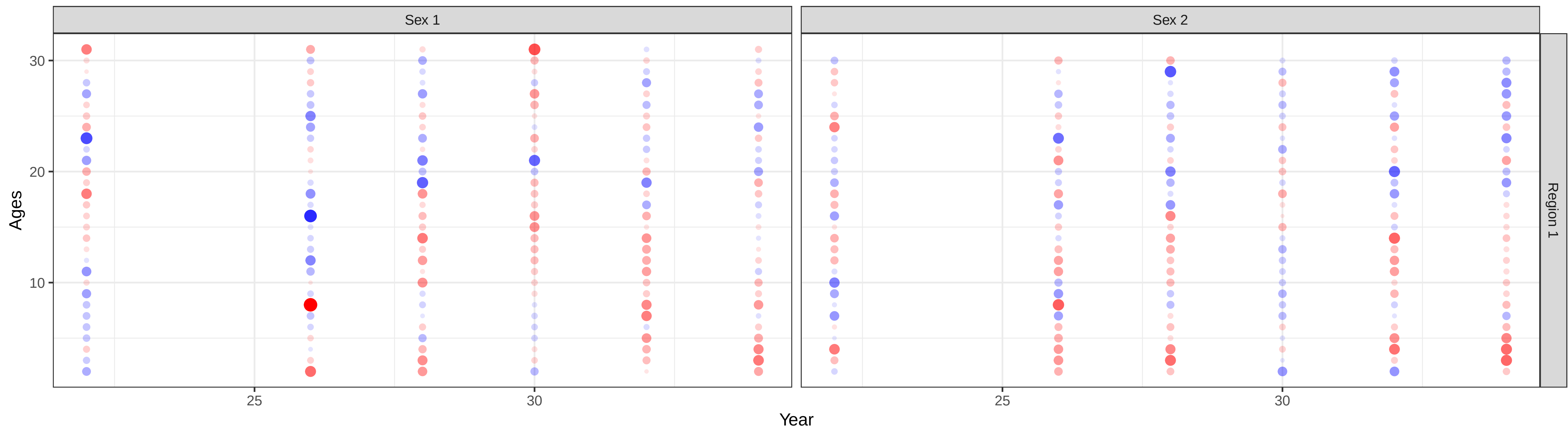




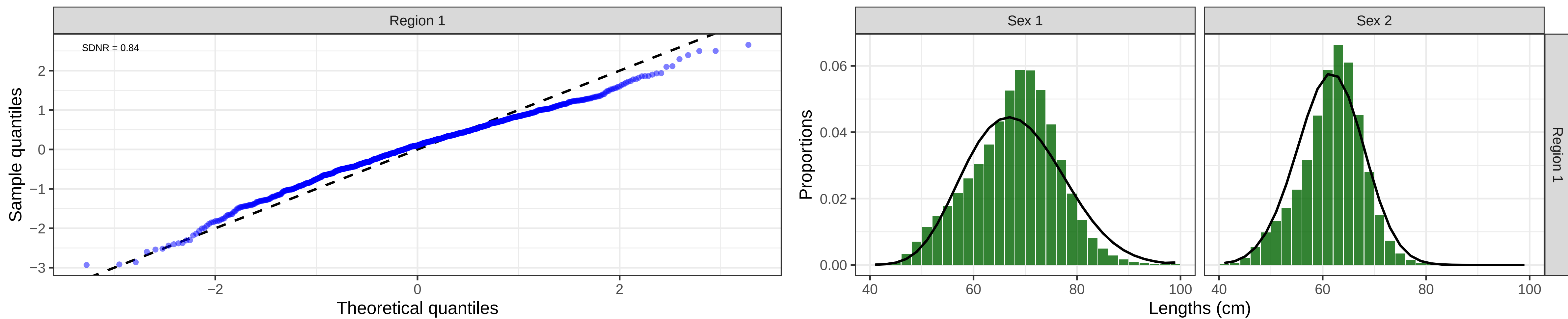
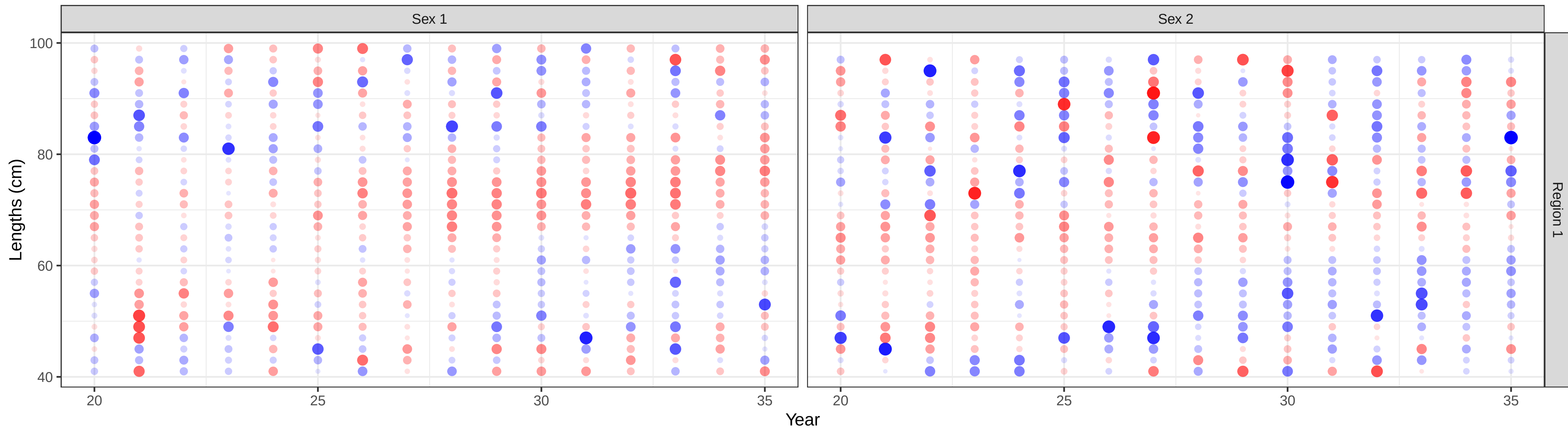
Sign ● Neg ● Pos abs(Resid) ● 1 ● 2 ● 3 ● 4



Sign ● Neg ● Pos abs(Resid) ● 1 ● 2 ● 3 ● 4

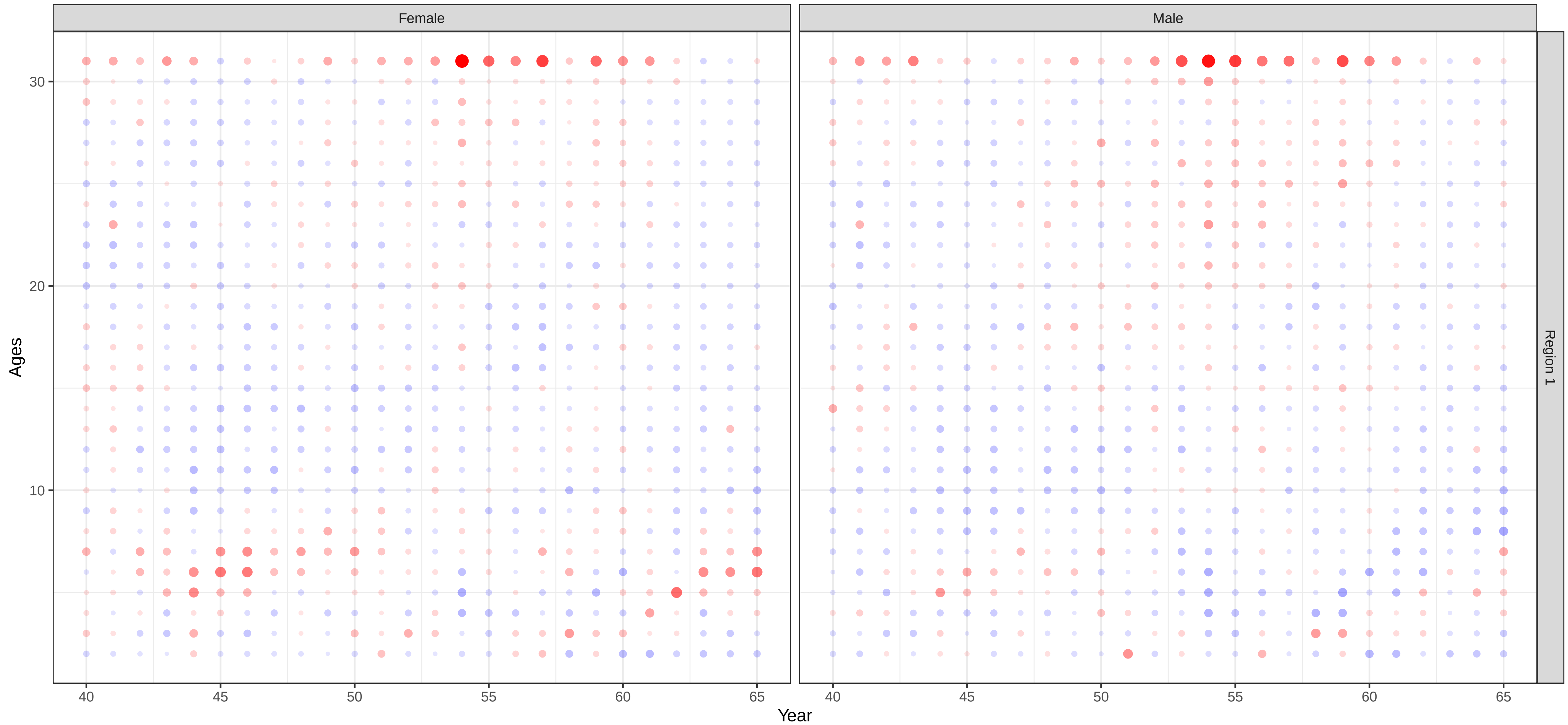


Sign ● Neg ● Pos abs(Resid) ● 1 ● 2



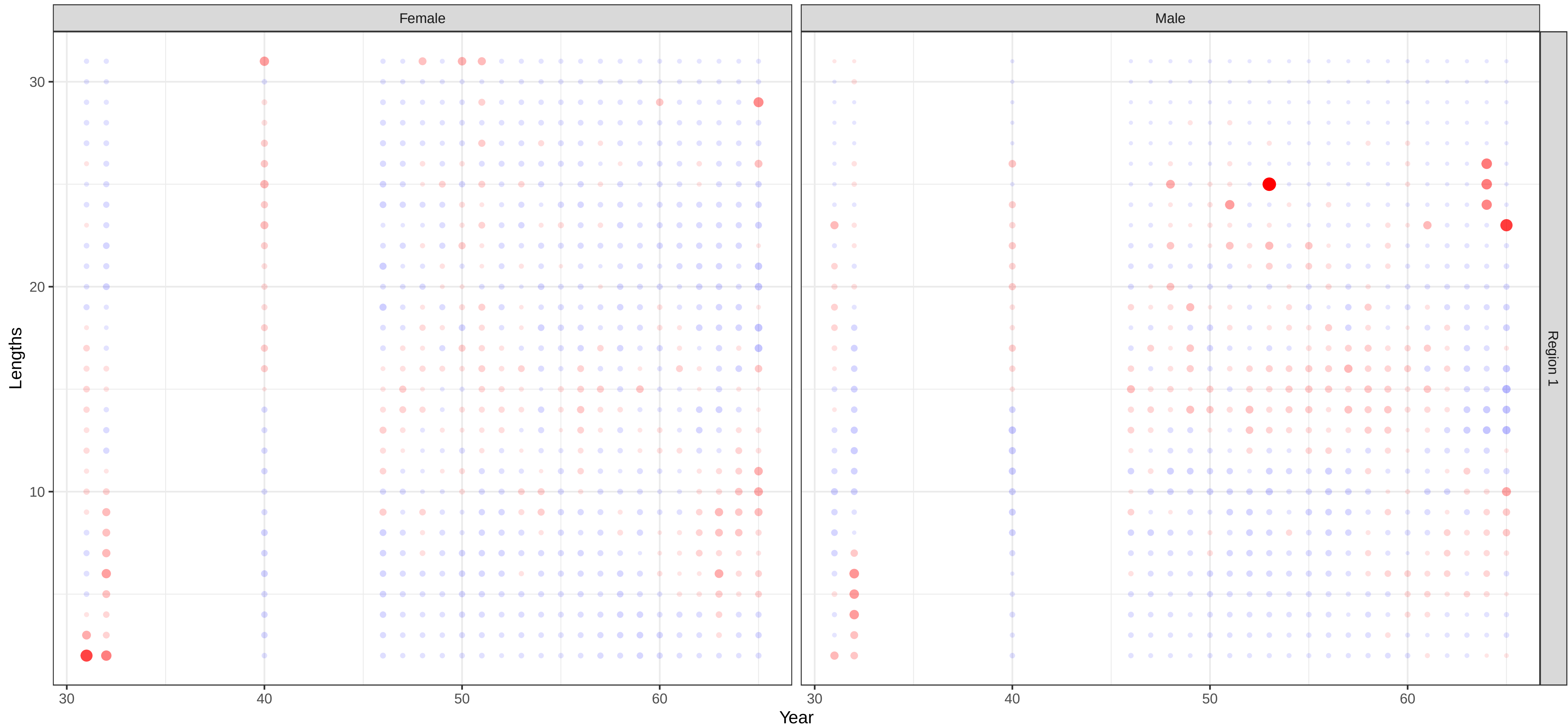
Fixed Gear Fishery Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 2 ● 4 ● 6



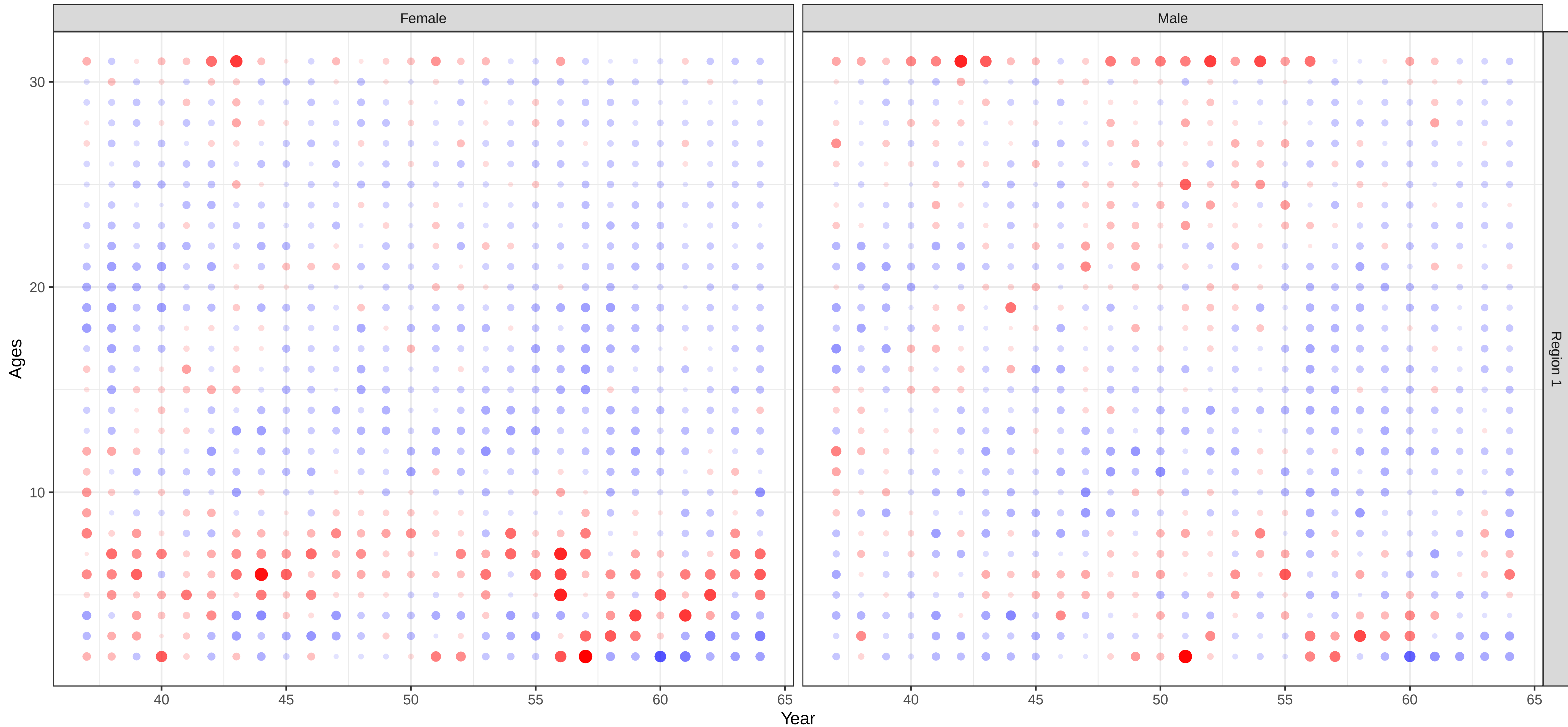
Trawl Fishery Length Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.0 ● 2.5 ● 5.0 ● 7.5



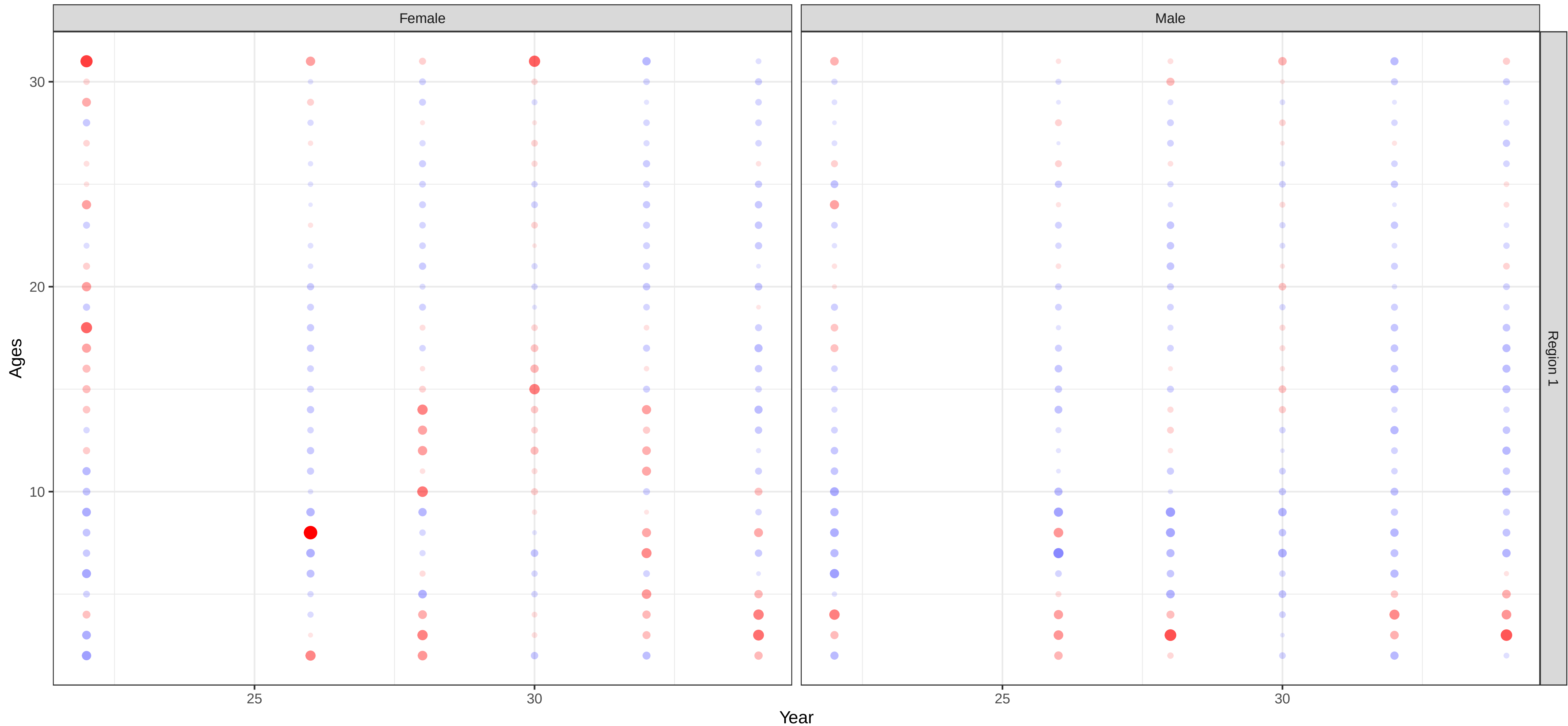
LLS Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 1 ● 2 ● 3 ● 4



JPN LLS Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 1 ● 2 ● 3 ● 4



JPN LLS Length Compositions

Sign ● Neg ● Pos abs(Resid) ● 0 ● 1 ● 2

